

Vacuum Moduli Spaces of Four-Dimensional $\mathcal{N} = 2$ SCFTs

A self-contained guide from protected operators to low-energy geometry

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Version July 13, 2026

Prepared with assistance from OpenAI Codex, based on the author's discussions with GPT-5.6 Sol.

Abstract

These notes explain how the vacuum branches of a four-dimensional $\mathcal{N} = 2$ superconformal field theory arise from protected local operators and how their geometry controls, but does not by itself exhaust, the low-energy physics. No Lagrangian presentation is assumed.

The discussion deliberately separates three layers that are often conflated: the protected operator ring, the algebraic geometry of its vacuum characters, and the additional dynamical data of the low-energy theory. This separation clarifies why a physical vacuum is a closed point rather than the whole spectrum, why $\text{Spec } \mathcal{R}_{\text{CB}}$ alone does not produce photons or Seiberg–Witten periods, and why a free tangent sector need not be the complete infrared theory. Within the Coulomb-branch low-energy layer, it further distinguishes the lattice of possible charges from the occupied BPS spectrum, separates the stable node particles from the composite representations built from them, and states precisely when a finite-length categorical heart supplies a noncircular proof of the positive-basis property.

The intended use is both linear reading and later lookup. Definitions state the precise content; intuition and reasoning boxes record why the definitions are natural and which extra assumptions enter each implication.

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Reader's Guide

The central question

Given the local operator data of an $\mathcal{N} = 2$ SCFT, which protected expectation values label supersymmetric vacua, what geometry do those vacua form, and what massless theory describes fluctuations around a chosen point?

The answer has three logically distinct layers:

Layer	Basic object	What it determines
Operator algebra	a protected ring $\mathcal{R}$	allowed holomorphic order parameters and their relations
Vacuum geometry	characters $\mathcal{R} \rightarrow \mathbb{C}$ and $\text{Spec } \mathcal{R}$	ordinary vacua, local rings, tangent directions, and singularities
Low-energy dynamics	massless multiplets, charge lattices, and effective couplings	metrics, central charges, duality monodromies, and the strict IR fixed point

How to use the note

- For a first pass, read [sections 2 to 4](#) and [6](#) and the final reasoning map in [section 8](#).
- To recover notation quickly, use the conceptual dictionary in [section 7](#).
- For Coulomb special geometry, start at [sections 3.3, 3.4](#) and [3.6](#).
- For massive BPS particles, chambers, quivers, finite-length hearts, and mutation, start at [sections 3.8](#) to [3.12](#).
- For the $\widehat{\mathcal{B}}_R$ notation and flavor moment maps, start at [sections 4.1](#) and [4.3](#).

The two implications that must never be silently reversed

physical vacuum $\implies$ character of the protected ring

does not, without an additional completeness assertion, prove that every algebraic character is physically realized. Likewise,

branch tangent modes become free in the strict IR

does not prove that the complete infrared theory contains no residual sector.

Throughout, the theory is assumed unitary and four-dimensional, and “flavor symmetry” means a continuous internal symmetry commuting with the full $\mathcal{N} = 2$ superconformal algebra. Signs and normalizations of $U(1)_r$ charge and of the central charge vary in the literature; invariant statements are emphasized whenever possible.

1 What Data Constitute a Four-Dimensional $\mathcal{N} = 2$ SCFT?

Purpose of this section

This section fixes the minimum common language used later: states, local operators, OPE data, symmetries, and superconformal multiplets. Readers already comfortable with radial quantization and $\mathfrak{su}(2, 2|2)$ may skip directly to [section 2](#).

1.1 A preliminary caveat

There is no single completely rigorous mathematical definition that presently covers every interacting four-dimensional quantum field theory. For the present purpose, it is therefore useful to describe a QFT by the physical data from which its observables and consistency conditions are determined.

Definition 1.1 (Quantum field theory data). A local quantum field theory on four-dimensional Minkowski space is characterized operationally by the following mutually compatible structures:

- (i) a space of quantum states;
- (ii) a collection of local operators;
- (iii) correlation functions of these operators;
- (iv) an operator product expansion encoding short-distance products;
- (v) a unitary action of spacetime and internal symmetries;
- (vi) locality, unitarity, and positivity conditions;
- (vii) a prescription for selecting a vacuum and constructing excitations above it.

The individual terms in this definition are explained below.

1.2 States and the Hilbert space

Definition 1.2 (Hilbert space). The *Hilbert space* $\mathcal{H}_{\text{phys}}$ is the complex vector space of physical quantum states equipped with a positive-definite inner product

$$\langle \psi | \chi \rangle.$$

A normalized state satisfies $\langle \psi | \psi \rangle = 1$.

The inner product determines probabilities. If a state is expanded in an orthonormal basis as

$$|\psi\rangle = \sum_n c_n |n\rangle,$$

then $|c_n|^2$ is the probability of obtaining the outcome associated with $|n\rangle$.

Definition 1.3 (Observable). An *observable* is represented by a self-adjoint operator acting on $\mathcal{H}_{\text{phys}}$. Its expectation value in a normalized state $|\psi\rangle$ is

$$\langle A \rangle_\psi = \langle \psi | A | \psi \rangle.$$

Definition 1.4 (Unitarity). A quantum theory is *unitary* if time evolution preserves the Hilbert-space inner product. If $U(t)$ is the time-evolution operator, then

$$U(t)^\dagger U(t) = \text{id}.$$

Unitarity ensures conservation of total probability and requires physical states to have nonnegative norm.

Remark 1.5. In relativistic QFT, the most basic objects are usually local operator-valued distributions rather than ordinary bounded operators. Their products must be interpreted inside correlation functions or after smearing with test functions.

1.3 Local operators and locality

Definition 1.6 (Local operator). A *local operator* $\mathcal{O}(x)$ is an operator inserted at a spacetime point x . It probes the theory in an arbitrarily small neighborhood of x .

Examples in a Lagrangian theory include a scalar field $\phi(x)$, the field strength $F_{\mu\nu}(x)$, and gauge-invariant composite operators such as $\text{Tr } \phi^k(x)$.

Definition 1.7 (Locality). A relativistic QFT is *local* if observables inserted at spacelike-separated points cannot influence one another. For bosonic operators this is expressed by

$$[\mathcal{O}_1(x), \mathcal{O}_2(y)] = 0, \quad (x - y)^2 > 0,$$

with an analogous graded commutator for fermionic operators.

Definition 1.8 (Lorentz scalar). A *Lorentz scalar operator* is unchanged by Lorentz transformations apart from the transformation of its insertion point. It carries Lorentz spins

$$(j_1, j_2) = (0, 0).$$

A constant nonzero VEV can preserve Lorentz invariance only when it belongs to a scalar operator.

Locality is what permits one to speak about independent measurements in causally disconnected regions.

1.4 Correlation functions

Definition 1.9 (Correlation function). Given a vacuum $|\Omega\rangle$, an n -point correlation function is

$$\langle O_1(x_1) \cdots O_n(x_n) \rangle := \langle \Omega | O_1(x_1) \cdots O_n(x_n) | \Omega \rangle,$$

with an appropriate time-ordering or Euclidean prescription.

Correlation functions contain the measurable dynamical information of the theory. Their singularities describe short-distance propagation, while their long-distance behavior describes the spectrum of low-energy states.

Definition 1.10 (Connected correlator). A *connected correlator* is the part of a correlation function that does not factorize into products of correlators involving disjoint subsets of insertions. For example,

$$\langle O_1 O_2 \rangle_{\text{conn}} = \langle O_1 O_2 \rangle - \langle O_1 \rangle \langle O_2 \rangle.$$

Definition 1.11 (Cluster decomposition). Let p be a pure vacuum, meaning a vacuum in a single superselection sector rather than a coherent superposition or statistical mixture of distinct vacua. The vacuum satisfies *cluster decomposition* if, for local operators at large spacelike separation,

$$\lim_{|\mathbf{x}-\mathbf{y}| \rightarrow \infty} \langle O_1(t, \mathbf{x}) O_2(t, \mathbf{y}) \rangle_p = \langle O_1 \rangle_p \langle O_2 \rangle_p.$$

Equivalently,

$$\lim_{|\mathbf{x}-\mathbf{y}| \rightarrow \infty} \langle O_1(t, \mathbf{x}) O_2(t, \mathbf{y}) \rangle_{p, \text{conn}} = 0.$$

Cluster decomposition expresses the statistical independence of experiments separated by an arbitrarily large distance within one pure vacuum. It is a long-distance statement; by itself it says nothing about the coincident-point product of two operators.

1.5 The operator product expansion

Definition 1.12 (Operator product expansion). The *operator product expansion*, or OPE, expresses the product of two nearby local operators as a sum of local operators:

$$O_i(x) O_j(0) \sim \sum_k C_{ij}^k(x) O_k(0), \quad x \rightarrow 0.$$

The functions $C_{ij}^k(x)$ are called *OPE coefficients* or *Wilson coefficients*.

The symbol $\sim$ means equality inside correlation functions as an asymptotic expansion at short distance. In a conformal field theory, the coordinate dependence of $C_{ij}^k(x)$ is strongly constrained, and the remaining numerical coefficients are part of the conformal data.

Remark 1.13. Associativity of the OPE means that a product of several operators gives the same result regardless of the order in which pairwise OPEs are performed. In four-point functions, this becomes the crossing-symmetry constraint.

1.6 Symmetries, charges, and conserved currents

Definition 1.14 (Symmetry). A *symmetry* is a transformation acting on states and operators that leaves all physical predictions invariant. Infinitesimally, a continuous symmetry is generated by a conserved charge Q .

Definition 1.15 (Spacetime and internal symmetries). A *spacetime symmetry* acts on spacetime coordinates and therefore on the placement of operators. An *internal symmetry* acts on internal labels of states and operators while leaving spacetime points fixed.

Definition 1.16 (Global symmetry and gauge redundancy). A *global symmetry* acts nontrivially on physical states and gauge-invariant operators. A *gauge symmetry* is instead a redundancy of a particular description: gauge-related field configurations represent the same physical configuration. Consequently, local physical operators must be gauge invariant.

For an operator O , the infinitesimal action of the symmetry is represented schematically by

$$\delta O = i[Q, O].$$

Definition 1.17 (Conserved current). A *conserved current* is a local vector operator $J^\mu(x)$ obeying

$$\partial_\mu J^\mu = 0.$$

The corresponding conserved charge is

$$Q = \int_{\mathbb{R}^3} d^3x J^0(x).$$

Definition 1.18 (Stress-energy tensor). The *stress-energy tensor* $T_{\mu\nu}$ is the conserved current associated with spacetime translations. It obeys

$$\partial^\mu T_{\mu\nu} = 0.$$

Its conserved charges are the four-momentum operators P_ν .

Definition 1.19 (Flavor symmetry). A *flavor symmetry* is a global internal symmetry that commutes with the entire superconformal algebra. Its group will be denoted by F . Flavor transformations act nontrivially on local operators but do not rotate the supercharges.

This distinguishes flavor symmetry from R-symmetry, which acts nontrivially on the supercharges themselves.

Definition 1.20 (Poincaré symmetry). *Poincaré symmetry* is the symmetry generated by translations P_μ and Lorentz transformations $M_{\mu\nu}$. It is the spacetime symmetry of a relativistic theory on flat Minkowski space.

Definition 1.21 (Conformal field theory). A *conformal field theory*, or CFT, is a local QFT invariant under the conformal group, which extends Poincaré transformations by dilatations and special conformal transformations.

Definition 1.22 (Supersymmetric field theory). A *supersymmetric field theory* has fermionic symmetry generators called supercharges. Their anticommutators contain spacetime translations, so supersymmetry relates bosonic and fermionic operators and states.

Definition 1.23 (Superconformal field theory). A *superconformal field theory*, or SCFT, is simultaneously conformal and supersymmetric, with both sets of generators combined into a superconformal algebra.

1.7 Conformal symmetry

The bosonic conformal algebra in four dimensions is $\mathfrak{so}(4, 2)$. Its generators are

$$P_\mu, \quad M_{\mu\nu}, \quad D, \quad K_\mu,$$

representing translations, Lorentz transformations, dilatations, and special conformal transformations.

Definition 1.24 (Scaling dimension). A local operator $\mathcal{O}(0)$ has *scaling dimension* Δ if

$$[D, \mathcal{O}(0)] = i\Delta \mathcal{O}(0)$$

in Lorentzian conventions. Equivalently, under a scale transformation $x \mapsto \lambda x$,

$$\mathcal{O}(x) \mapsto \lambda^{-\Delta} \mathcal{O}(\lambda x).$$

Definition 1.25 (Conformal primary and descendant). A *conformal primary operator* is annihilated at the origin by all special conformal generators:

$$[K_\mu, \mathcal{O}(0)] = 0.$$

Operators obtained by acting with translations P_μ are called *conformal descendants*.

1.8 Supersymmetry and the $\mathcal{N} = 2$ superconformal algebra

Four-dimensional $\mathcal{N} = 2$ Poincaré supersymmetry contains eight real supercharges, conventionally written as

$$Q_\alpha^I, \quad \tilde{Q}_{I\dot{\alpha}}, \quad I = 1, 2.$$

Here α and $\dot{\alpha}$ are left- and right-handed Lorentz spinor indices, while I is an $SU(2)_R$ index.

The full 4d $\mathcal{N} = 2$ superconformal algebra is $\mathfrak{su}(2, 2|2)$. Its bosonic subalgebra is

$$\mathfrak{so}(4, 2) \oplus \mathfrak{su}(2)_R \oplus \mathfrak{u}(1)_r.$$

The last two factors form the R-symmetry algebra.

The notation and shortening terminology used below follow the systematic multiplet classification of Córdova, Dumitrescu, and Intriligator [1].

Definition 1.26 (R-symmetry). An *R-symmetry* is an internal symmetry that acts nontrivially on the supercharges. For 4d $\mathcal{N} = 2$ theories, the connected R-symmetry group is locally

$$SU(2)_R \times U(1)_r.$$

We denote the $SU(2)_R$ spin of an operator by

$$R = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots$$

The corresponding irreducible representation has dimension $2R + 1$. The normalization and sign of the $U(1)_r$ charge differ across the literature, so Coulomb-branch operators will primarily be labeled by their scaling dimensions Δ .

Definition 1.27 (Superconformal primary). A *superconformal primary* is annihilated at the origin by the special conformal generators K_μ and by the conformal supercharges S and $\tilde{S}$.

Definition 1.28 (Superconformal multiplet). A *superconformal multiplet* is the collection of operators generated from one superconformal primary by repeated action of the Poincaré supercharges $Q, \tilde{Q}$ and translations P_μ .

Definition 1.29 (Long and short multiplets). A *long multiplet* has no additional null relations beyond those required by the algebra. A *short multiplet* satisfies extra annihilation conditions because its quantum numbers saturate a unitarity bound. Such shortened multiplets are often called *BPS multiplets*.

Definition 1.30 (Protected scalar operator). In these notes, a *protected scalar operator* means a specified Lorentz-scalar superconformal primary that obeys specified supercharge-annihilation conditions and whose scaling dimension is fixed by the corresponding shortening relation. The annihilating supercharges and the other quantum numbers are part of the definition; they differ between the Coulomb and Higgs sectors.

Definition 1.31 (Protected operator ring). A *protected operator ring* $\mathcal{R}$ is a unital complex vector space of protected scalar operators, or their protected-cohomology classes, equipped with a product

$$O_1 \cdot O_2 \in \mathcal{R}$$

obtained from the regular protected term in their OPE. The product is required to close in the chosen protected sector and to be associative and commutative after the relevant cohomological or contact-term identifications.

The phrase “BPS operator” will only be used as shorthand for such a specified protected primary. A *BPS multiplet* contains the primary together with many supercharge descendants, and those descendants are not all scalar order parameters or moduli-space coordinates. Thus “this operator generates a BPS multiplet” and “every operator in that multiplet is a branch coordinate” are very different statements.

BPS shortening protects certain operator dimensions and permits the intrinsic identification of Coulomb- and Higgs-branch primaries even when no Lagrangian is known. Their precise shortening conditions will be given before their rings are defined.

1.9 The operator spectrum and conformal data

Definition 1.32 (Operator spectrum). The *operator spectrum* is the collection of local operators, organized into irreducible superconformal multiplets and labeled by their quantum numbers.

For a superconformal primary, the relevant labels include

$$(\Delta; j_1, j_2; R; r; \rho_F),$$

where:

- Δ is the scaling dimension;

- (j_1, j_2) are the spins under the Lorentz group $SU(2)_1 \times SU(2)_2$;
- R is the $SU(2)_R$ spin;
- r is the $U(1)_r$ charge;
- ρ_F is a representation of the flavor group F .

Definition 1.33 (Conformal data). The *conformal data* consist of the spectrum of conformal primary operators together with their OPE coefficients. In a unitary CFT, these data are constrained by unitarity, locality, OPE associativity, and crossing symmetry.

In this operational sense, specifying all consistent conformal data specifies the local observables of the CFT.

1.10 State–operator correspondence

Definition 1.34 (State–operator correspondence). In a CFT, radial quantization identifies a local operator inserted at the origin with a state on the three-sphere:

$$\mathcal{O}(0)|0\rangle \longleftrightarrow |O\rangle \in \mathcal{H}_{S^3}.$$

The scaling dimension of O equals the energy of the corresponding state on S^3 .

This correspondence allows statements about local operators to be translated into statements about states and superconformal representations.

2 Vacua and Vacuum Moduli Spaces

From a state to a point of geometry

The key task is to justify the chain

$$\begin{array}{ccc} \text{supersymmetric vacuum} & \longrightarrow & \text{VEVs of protected operators} \\ & \Downarrow & \\ \text{character of a protected ring} & \longrightarrow & \text{closed point of an affine spectrum.} \end{array}$$

The Ward-identity and cluster-decomposition steps in the middle are essential; multiplicativity is not automatic for arbitrary local operators.

2.1 Vacuum states

Definition 2.1 (Poincaré-invariant vacuum). A *Poincaré-invariant vacuum* is a state $|\Omega\rangle$ satisfying

$$P_\mu |\Omega\rangle = 0, \quad M_{\mu\nu} |\Omega\rangle = 0.$$

It is translationally and Lorentz invariant.

Definition 2.2 ($\mathcal{N} = 2$ supersymmetric vacuum). A Poincaré-invariant vacuum is an $\mathcal{N} = 2$ supersymmetric vacuum if it is annihilated by all eight real Poincaré supercharges:

$$Q_\alpha^I |\Omega\rangle = 0, \quad \tilde{Q}_{I\dot{\alpha}} |\Omega\rangle = 0.$$

Definition 2.3 (Superconformal vacuum). A superconformal vacuum is invariant under the full superconformal group. In particular, it is invariant under dilatations, special conformal transformations, and conformal supercharges in addition to Poincaré supersymmetry.

For an SCFT with a moduli space, the superconformal vacuum is the distinguished point at which all positive-dimension order parameters vanish. It is usually called the *origin* or the *tip of the cone*.

2.2 Vacuum expectation values

Definition 2.4 (Vacuum expectation value). The *vacuum expectation value*, or VEV, of a local operator $\mathcal{O}$ in a vacuum p is

$$\langle \mathcal{O} \rangle_p := \langle \Omega_p | \mathcal{O} | \Omega_p \rangle.$$

Translation invariance implies that the VEV of a local operator is independent of its insertion point. Lorentz invariance implies that a nonzero VEV can be assigned only to a Lorentz scalar, unless Lorentz symmetry is spontaneously broken.

Remark 2.5. The statement that a vacuum preserves supersymmetry is not identical to the statement that every operator with a nonzero VEV is annihilated by every supercharge. Rather, the vacuum state itself is annihilated by all Poincaré supercharges, while the operators whose VEVs parameterize supersymmetric vacua lie in suitable short multiplets.

2.3 The vacuum moduli space

Definition 2.6 (Vacuum moduli space). The *vacuum moduli space* $\mathcal{M}_{\text{vac}}$ is the space of inequivalent Poincaré-invariant vacua that minimize the energy. The $\mathcal{N} = 2$ supersymmetric vacuum moduli space consists of those vacua that preserve all eight Poincaré supercharges.

The word *modulus* means a continuously variable parameter labeling inequivalent vacua. A *branch* is a connected or irreducible component, or more generally a geometrically distinguished stratum, of the vacuum moduli space.

Definition 2.7 (Order parameter). An *order parameter* is an operator whose VEV distinguishes different vacua. On a supersymmetric moduli space, specified protected scalar primaries can serve as holomorphic order parameters.

Suppose a set of scalar operators $\mathcal{O}_i$ obeys algebraic relations

$$f_a(\mathcal{O}_1, \dots, \mathcal{O}_n) = 0.$$

Their VEVs

$$u_i = \langle \mathcal{O}_i \rangle$$

must satisfy

$$f_a(u_1, \dots, u_n) = 0.$$

Thus operator relations become equations defining the vacuum geometry.

For the branch chiral rings used below, supersymmetric Ward identities imply a further property. Schematically, translations of a protected insertion are exact with respect to a supercharge that annihilates both the insertions and the supersymmetric vacuum,

$$\partial_\mu \mathcal{O}(x) = \{Q, \Lambda_\mu(x)\}.$$

Consequently, separated correlators of compatible protected insertions are independent of their separation. This is the missing bridge between the coincident protected OPE product and the long-distance statement of cluster decomposition. It is not true for arbitrary QFT operators.

Proposition 2.8 (Vacua as characters of an operator ring). *Let $\mathcal{R}$ be a protected operator ring for which the preceding Ward-identity position independence holds, and let p be a normalized pure supersymmetric vacuum satisfying cluster decomposition. Then*

$$\chi_p : \mathcal{R} \longrightarrow \mathbb{C}, \quad \chi_p(\mathcal{O}) = \langle \mathcal{O} \rangle_p.$$

is a unital $\mathbb{C}$ -algebra homomorphism. In particular,

$$\chi_p(\mathcal{O}_1 \cdot \mathcal{O}_2) = \chi_p(\mathcal{O}_1) \chi_p(\mathcal{O}_2).$$

Proof. Linearity of expectation values gives $\mathbb{C}$ -linearity, and normalization gives

$$\chi_p(1) = 1.$$

For two protected operators, the definition of the protected OPE product and the Ward identity allow the insertions to be separated without changing the correlator. Therefore

$$\begin{aligned} \chi_p(\mathcal{O}_1 \cdot \mathcal{O}_2) &= \langle \mathcal{O}_1(x) \mathcal{O}_2(y) \rangle_p \\ &= \lim_{|x-y| \rightarrow \infty} \langle \mathcal{O}_1(x) \mathcal{O}_2(y) \rangle_p \\ &= \langle \mathcal{O}_1 \rangle_p \langle \mathcal{O}_2 \rangle_p, \end{aligned}$$

where the final equality is cluster decomposition. □

The kernel

$$\mathfrak{m}_p := \ker \chi_p$$

is a maximal ideal, since the unital $\mathbb{C}$ -algebra map has image $\mathbb{C}$ and hence

$$\mathcal{R}/\mathfrak{m}_p \simeq \mathbb{C}.$$

Thus a complex-valued physical vacuum determines a closed $\mathbb{C}$ -point, not an arbitrary prime ideal.

Remark 2.9. This proposition proves the direction

$$\text{physical vacuum} \longrightarrow \text{character of the protected ring}.$$

The converse claim that every algebraic character is realized by a physical vacuum is an additional physical assertion implicit in identifying the complete branch with the ring spectrum.

Why the VEV map is multiplicative

$$\begin{aligned}\chi_p(\mathcal{O}_1\mathcal{O}_2) &= \langle \mathcal{O}_1(x)\mathcal{O}_2(y) \rangle_p && \text{protected OPE and position independence,} \\ &= \lim_{|x-y|\rightarrow\infty} \langle \mathcal{O}_1(x)\mathcal{O}_2(y) \rangle_p && \text{move the insertions apart,} \\ &= \langle \mathcal{O}_1 \rangle_p \langle \mathcal{O}_2 \rangle_p && \text{cluster decomposition.}\end{aligned}$$

The first equality is the supersymmetric input; the last is the vacuum-sector input.

2.4 Ordinary vacua, maximal ideals, and the affine spectrum

Definition 2.10 (Affine spectrum and complex points). For a commutative ring R ,

$$\text{Spec } R = \{\mathfrak{p} \subset R : \mathfrak{p} \text{ is prime}\}, \quad \text{MaxSpec } R = \{\mathfrak{m} \subset R : \mathfrak{m} \text{ is maximal}\}.$$

If R is a $\mathbb{C}$ -algebra, its ordinary complex points are

$$(\text{Spec } R)(\mathbb{C}) := \text{Hom}_{\mathbb{C}\text{-alg}}(R, \mathbb{C}).$$

For a finitely generated $\mathbb{C}$ -algebra, Hilbert's Nullstellensatz identifies these complex points with $\text{MaxSpec } R$.

Accordingly, a single vacuum is neither the whole branch nor the whole spectrum. It corresponds to one maximal ideal $\mathfrak{m}_p$. The notation $\text{Spec } R$ packages the ordinary closed points together with the algebraic functions and their local behavior.

For example,

$$\text{Spec } \mathbb{C}[u]$$

contains the closed points $(u - a)$, $a \in \mathbb{C}$, as well as the non-closed prime (0) . The latter is the generic point of the whole irreducible affine line; it is not an extra vacuum with a definite complex value of u .

The most useful extra structure is often the local ring

$$\mathcal{O}_{\text{Spec } R, p} = R_{\mathfrak{m}_p},$$

which records holomorphic functions and relations near the vacuum. It determines, for example, the Zariski tangent space

$$T_p \text{Spec } R \simeq \text{Hom}_{\mathbb{C}}(\mathfrak{m}_p/\mathfrak{m}_p^2, \mathbb{C})$$

and helps distinguish smooth from singular points.

The scheme can also remember nilpotent information invisible to ordinary complex points. The rings

$$\mathbb{C}[u] \quad \text{and} \quad \mathbb{C}[u, \varepsilon]/(\varepsilon^2)$$

have the same complex-valued points, because every character sends ε to zero, but different local rings and tangent spaces. Whether such a nonreduced structure carries physical information must be decided theory by theory; many familiar Coulomb chiral rings are assumed to be reduced, and often freely generated.

Thus, if one only wants the set of ordinary complex vacua, $\text{MaxSpec } R$ or $\text{Hom}_{\mathbb{C}\text{-alg}}(R, \mathbb{C})$ is enough. The standard notation $\text{Spec } R$ is used when one also wants generic subvarieties, local rings, multiplicities, infinitesimal structure, and functorial operations such as quotients and localization.

Why keep Spec R ?

The non-closed points are not extra physical vacua. They are bookkeeping devices for generic behavior on irreducible subvarieties. The genuinely useful gain is the attached local algebra: it remembers which functions exist near a vacuum, how branches meet, whether the point is smooth, and whether infinitesimal or nilpotent structure is present.

2.5 Spontaneous breaking of conformal symmetry

Let $\mathcal{O}$ be a scalar operator of positive scaling dimension Δ , and suppose

$$\langle \mathcal{O} \rangle = v \neq 0.$$

The VEV defines a physical scale

$$M_v \sim |v|^{1/\Delta}.$$

A nonzero scale is incompatible with an invariant conformal vacuum. Therefore, a generic nonzero VEV spontaneously breaks

$$D, \quad K_\mu, \quad S, \quad \tilde{S},$$

while it may preserve

$$P_\mu, \quad M_{\mu\nu}, \quad Q, \quad \tilde{Q}.$$

Thus one should distinguish

the SCFT as a theory from a particular vacuum of that theory.

The theory possesses superconformal symmetry, but a nonzero-VEV vacuum generally preserves only $\mathcal{N} = 2$ Poincaré supersymmetry.

3 The Coulomb Branch

What is intrinsic and what is additional

The Coulomb chiral ring intrinsically defines an affine complex geometry. The $U(1)^r$ photons, electromagnetic lattice, central charge, metric, and Seiberg–Witten periods arise only after adding the low-energy $\mathcal{N} = 2$ dynamics on the physical regular locus. Even those data specify only possible charge sectors and their BPS bounds; the occupied BPS spectrum, its chambers, and any quiver presentation require further dynamical and stability data.

3.1 Coulomb-branch operators

Definition 3.1 (Coulomb-branch chiral primary). A *Coulomb-branch chiral primary* is a superconformal primary $\mathcal{O}(0)$ satisfying

$$j_1 = j_2 = 0, \quad R = 0,$$

and annihilated by all four Poincaré supercharges of one chirality, which we conventionally take to be

$$\tilde{Q}_{I\dot{\alpha}}\mathcal{O}(0) = 0, \quad I = 1, 2, \quad \dot{\alpha} = 1, 2.$$

The shortening relation fixes its scaling dimension in terms of its nonzero $U(1)_r$ charge, with a sign and normalization that depend on convention.

The short multiplet generated from such a primary is conventionally called an $\mathcal{E}$ -type or $\bar{\mathcal{E}}$ -type multiplet, depending on which chirality and $U(1)_r$ convention is used [1]. In these notes, “ $\mathcal{E}$ -type” is only a classification label for the explicit conditions above; it is not an additional ingredient in the definition.

Remark 3.2. The Coulomb-branch coordinate is the bottom scalar primary $\mathcal{O}$, not the entire $\mathcal{E}$ -type multiplet. Its fermionic and higher descendants are indispensable members of the supersymmetry representation but are not independent Lorentz-invariant vacuum coordinates.

Primary versus multiplet

A multiplet is the supersymmetry orbit generated from one primary. A vacuum coordinate must itself be a bosonic Lorentz scalar with a constant VEV. The primary has precisely this role; most descendants have spin, fermion parity, or derivative structure that prevents them from being independent Lorentz-invariant order parameters.

Proposition 3.3 (Flavor neutrality of Coulomb primaries). *Every Coulomb-branch chiral primary is a singlet under every continuous flavor symmetry.*

Proof. Let $\mathcal{O}_A$ transform in a putative flavor representation,

$$[F^a, \mathcal{O}_A] = (T^a)_A{}^B \mathcal{O}_B.$$

Choose an $\mathcal{N} = 1$ subalgebra contained in the $\mathcal{N} = 2$ algebra. The Coulomb primary is chiral for this subalgebra. The $\mathcal{N} = 1$ flavor Ward identity says that a nonzero matrix T^a requires the scalar bottom component of the flavor-current multiplet to appear in the $\mathcal{O}_A \times \bar{\mathcal{O}}^B$ OPE with coefficient proportional to $(T^a)_A{}^B$.

In an $\mathcal{N} = 2$ SCFT, that scalar belongs to the moment-map triplet $\mu^{a(ij)}$, the superconformal primary of the $\widehat{\mathcal{B}}_1$ flavor-current multiplet described below. But $\mathcal{O}_A$ and $\bar{\mathcal{O}}^B$ are both $SU(2)_R$ singlets, whereas $\mu^{a(ij)}$ has $R = 1$. Hence

$$\langle \mathcal{O}_A \bar{\mathcal{O}}^B \mu^{a(ij)} \rangle = 0$$

by $SU(2)_R$ selection rules. The Ward identity therefore forces $T^a = 0$. □

This selection-rule viewpoint also underlies mixed Coulomb/moment-map correlator analyses such as [3].

The chirality assumption is essential in this argument: an arbitrary long-multiplet operator with $R = 0$ need not be flavor neutral. Continuous flavor transformations consequently do not move points of the pure Coulomb base. Flavor data can nevertheless affect Coulomb-branch physics through flavor-charged BPS states and background mass parameters in Z_γ .

The protected OPE product of two Coulomb-branch chiral primaries again has a Coulomb-branch chiral primary as its protected component, modulo protected-cohomology and operator relations.

Definition 3.4 (Coulomb-branch chiral ring). The *Coulomb-branch chiral ring* $\mathcal{R}_{\text{CB}}$ is the commutative ring generated by Coulomb-branch chiral primaries, modulo their operator relations.

Let u_i denote their VEVs:

$$u_i = \langle O_i \rangle.$$

Definition 3.5 (Coulomb branch). The *Coulomb branch* is a family of supersymmetric Poincaré-invariant vacua. It is parameterized by VEVs of Coulomb-branch chiral primaries, with Higgs-type order parameters set to zero. Its affine algebraic avatar is

$$\mathcal{C} = \text{Spec } \mathcal{R}_{\text{CB}}.$$

The ordinary complex-valued Coulomb vacua are the closed points

$$\mathcal{C}(\mathbb{C}) = \text{Hom}_{\mathbb{C}\text{-alg}}(\mathcal{R}_{\text{CB}}, \mathbb{C}) \simeq \text{MaxSpec } \mathcal{R}_{\text{CB}},$$

under the usual finite-generation assumption. A vacuum p corresponds to

$$\mathfrak{m}_p = \ker \left(\mathcal{O} \mapsto \langle \mathcal{O} \rangle_p \right).$$

The entire Coulomb branch is not one maximal ideal. Moreover, $\text{Spec } \mathcal{R}_{\text{CB}}$ encodes only the affine complex algebraic or scheme structure. It does not by itself imply the existence of photons, a metric, an electromagnetic charge lattice, or Seiberg–Witten periods; those arise from additional low-energy $\mathcal{N} = 2$ dynamics.

3.2 Basic geometric properties

Proposition 3.6. *The Coulomb branch is a complex cone with its superconformal vacuum at the tip. If u_i has scaling dimension Δ_i , the complexified scaling action is locally*

$$u_i \mapsto \lambda^{\Delta_i} u_i, \quad \lambda \in \mathbb{C}^*.$$

Definition 3.7 (Rank). The *rank* of a four-dimensional $\mathcal{N} = 2$ theory is the complex dimension of the Coulomb branch at a generic smooth point:

$$\text{rk}(\mathcal{T}) = \dim_{\mathbb{C}} \mathcal{C}.$$

This is the local dimension of the branch, not necessarily the number of displayed chiral-ring generators: generators can obey relations.

Definition 3.8 (Physical regular locus). The *physical regular locus* $\mathcal{C}^\circ \subset \mathcal{C}$ is the open region on which the charged low-energy sector has a nonsingular Abelian description, the vector-multiplet kinetic matrix is nondegenerate, and no additional electrically or magnetically charged state is massless. Its complement

$$\Delta_{\text{phys}} := \mathcal{C} \setminus \mathcal{C}^\circ$$

is the physical discriminant locus.

The physical regular locus must not be identified blindly with the scheme-theoretic smooth locus. A point of the base variety can be algebraically smooth while the special Kähler metric and Abelian effective action are singular because an extra charged state becomes massless.

3.3 Why rank r gives r Abelian vector multiplets

The relevant massless 4d $\mathcal{N} = 2$ Poincaré multiplets have different scalar content:

	bosonic fields	scalar moduli
vector multiplet	A_μ and one complex scalar φ	1 complex direction
hypermultiplet	4 real scalars	1 quaternionic direction

A hypermultiplet contains no photon.

Why does a tangent direction to a vacuum branch give a massless scalar? In a Lagrangian description, let $\Phi^A(t)$ be a curve of degenerate minima of the scalar potential. At a vacuum Φ_0 its tangent v^A obeys

$$v^A \left. \frac{\partial^2 V}{\partial \Phi^A \partial \Phi^B} \right|_{\Phi_0} v^B = 0.$$

The Hessian is the scalar mass-squared matrix, so the tangent fluctuation is massless. Intrinsically, the same conclusion follows by allowing a slowly varying vacuum parameter m^i to become a field $m^i(x)$: the effective action begins with derivative terms and has no potential along the moduli space.

The implication “modulus $\Rightarrow$ massless scalar” does not by itself decide which supersymmetry multiplet contains that scalar. That is extra $\mathcal{N} = 2$ representation-theoretic information. Coulomb-type directions, identified in the UV by the $\mathcal{E}$ -type shortening condition, become vector-multiplet scalar directions in the infrared; Higgs-type directions, identified by $\widehat{\mathcal{B}}_R$ shortening, become hypermultiplet directions.

Proposition 3.9 (Generic Coulomb multiplets). *Let $p \in C^\circ$ lie on a rank- r Coulomb branch. The low-energy theory contains exactly r massless Abelian vector multiplets associated with the Coulomb base. Consequently its local gauge algebra contains*

$$\mathfrak{u}(1)^r.$$

Indeed, $T_p C$ contains r independent complex Coulomb moduli, while one $\mathcal{N} = 2$ vector multiplet contains exactly one complex scalar. Each of the resulting r massless spin-one fields requires an Abelian gauge redundancy. At a physical regular point there are no extra massless charged vector bosons that would complete these photons into a non-Abelian algebra. Conversely, an additional massless vector multiplet would contain another freely variable complex scalar and would increase the local Coulomb rank.

This statement is a structural input from the low-energy realization of $\mathcal{N} = 2$ supersymmetry; it does not follow from the algebraic identity $C = \text{Spec } \mathcal{R}_{\text{CB}}$ alone.

Why the number is exactly r

$$\dim_{\mathbb{C}} T_p C = r \implies r \text{ massless complex Coulomb moduli} \implies r \text{ vector multiplets} \implies \mathfrak{u}(1)^r$$

at a physical regular point. Hypermultiplets may coexist, but they do not alter this count. An additional vector multiplet would bring an additional complex scalar modulus and hence increase the rank.

Remark 3.10 (Two different $U(1)$ s). The UV superconformal $U(1)_r$ is a global R-symmetry that rotates supercharges and Coulomb primaries. The $U(1)_{\text{gauge}}^r$ factors in the generic infrared theory are gauge

redundancies with photons. A Coulomb primary's $U(1)_r$ charge helps identify the type of order parameter; it does not generate or equal an infrared electric gauge charge. In fact, a nonzero Coulomb VEV generally breaks $U(1)_r$, while the scalar in an Abelian vector multiplet is neutral under its own Abelian gauge group.

The r vector multiplets need not exhaust every massless field. There can also be h neutral hypermultiplets over a generic Coulomb point, giving locally an enhanced Coulomb or mixed branch with a hyperkähler fiber. Charged hypermultiplets are instead generically massive and become massless only on special loci. Thus r counts vector multiplets associated with the Coulomb base, not all massless multiplets.

3.4 The $\mathcal{N} = 2$ Poincaré central charge

A nonzero Coulomb VEV breaks conformal symmetry but preserves $\mathcal{N} = 2$ Poincaré supersymmetry. The latter algebra admits a complex central operator $\widehat{Z}$. Up to convention-dependent constants and the placement of complex conjugation,

$$\{Q_\alpha^I, \widetilde{Q}_{J\dot{\beta}}\} = 2\delta^I_J \sigma_{\alpha\dot{\beta}}^\mu P_\mu,$$

and

$$\{Q_\alpha^I, Q_\beta^J\} = 2\epsilon_{\alpha\beta} \epsilon^{IJ} \widehat{Z}, \quad \{\widetilde{Q}_{I\dot{\alpha}}, \widetilde{Q}_{J\dot{\beta}}\} = 2\epsilon_{\dot{\alpha}\dot{\beta}} \epsilon_{IJ} \widehat{Z}.$$

It is central with respect to the super-Poincaré generators. It is unrelated to the Virasoro central charge of a two-dimensional CFT and to the four-dimensional Weyl-anomaly coefficients usually denoted a and c .

Central extensions of supersymmetry and their relation to BPS bounds were established in [4]; their exact realization on a four-dimensional $\mathcal{N} = 2$ Coulomb branch is central to the Seiberg–Witten solution [5, 6].

Fix a Coulomb vacuum p and a superselection sector of asymptotic states. Schur's lemma says only that $\widehat{Z}$ acts by a complex number on each irreducible super-Poincaré representation:

$$\widehat{Z}|\Psi\rangle = z|\Psi\rangle.$$

It does not, by itself, say what z is or that it is determined by electromagnetic charge.

The stronger statement comes from the concrete QFT realization of the supersymmetry algebra. The supercharges are spatial integrals of the supercurrent. Their anticommutator contains a total derivative, so the central operator is a surface charge at spatial infinity [4, 5]:

$$\widehat{Z} = \int_{S_\infty^2} (\text{scalar fields}) (\text{electric and magnetic fluxes}).$$

In a local Abelian frame this gives the operator identity, up to normalization,

$$\widehat{Z} = \sum_{I=1}^r (a^I(p) \widehat{n}_e^I + a_{D,I}(p) \widehat{n}_{m,I}) + \sum_A m^A \widehat{s}_A.$$

Here $\widehat{n}_e^I$ and $\widehat{n}_{m,I}$ measure electric and magnetic fluxes, while $\widehat{s}_A$ are flavor charges and m^A are background flavor-vector-multiplet masses. The flavor term is absent when the masses vanish.

The vacuum p enters because it fixes the scalar boundary conditions at infinity,

$$\varphi^I(x) \longrightarrow a^I(p) \quad (|x| \rightarrow \infty),$$

and the charge sector enters because it fixes the flux eigenvalues. Thus a state with full charge label

$$\gamma = (n_e^I, n_{m,I}, s_A)$$

has

$$Z_\gamma(p) = \sum_I (n_e^I a^I(p) + n_{m,I} a_{D,I}(p)) + \sum_A s_A m^A.$$

This is a consequence of a surface-charge operator relation, not merely of mutual commutativity. Two states with the same p and the same full charge γ can still have different energies, spins, particle content, and BPS properties; they nevertheless have the same central-charge eigenvalue.

Positivity of the supersymmetry anticommutator in the rest frame gives the BPS bound

$$M \geq |Z_\gamma|,$$

in a normalization in which the conventional proportionality factor is one. A BPS state saturates this bound and is annihilated by part of the supersymmetry algebra.

Why p and γ determine $Z_\gamma(p)$

$\widehat{Z}$ is central $\implies \widehat{Z}$ is scalar on an irreducible supermultiplet

is only Schur's lemma. The stronger field-theory input is

$$\widehat{Z} = a^I(p) \widehat{n}_{e,I} + a_{D,I}(p) \widehat{n}_m^I + m^A \widehat{s}_A.$$

The vacuum fixes the asymptotic scalar coefficients; the charge sector fixes the flux eigenvalues. Only after this boundary-charge identity is used does (p, γ) determine the eigenvalue.

3.5 Electromagnetic charge lattice and special coordinates

At each $p \in C^\circ$, the allowed electric and magnetic charges form an integral lattice

$$(\Gamma_{\text{em}})_p \simeq \mathbb{Z}^{2r}$$

with the antisymmetric Dirac pairing

$$\langle \cdot, \cdot \rangle : (\Gamma_{\text{em}})_p \times (\Gamma_{\text{em}})_p \longrightarrow \mathbb{Z}.$$

The lattices vary locally constantly over C° and therefore form an integral local system

$$\Gamma_{\text{em}} \longrightarrow C^\circ.$$

Parallel transport around a nontrivial loop can return a charge basis transformed by electric–magnetic monodromy.

On a simply connected patch, and assuming a principal polarization for simplicity, choose a symplectic basis

$$e_I, m^I, \quad \langle e_I, m^J \rangle = \delta_I^J, \quad \langle e_I, e_J \rangle = \langle m^I, m^J \rangle = 0.$$

More general global forms and line-operator spectra can modify the integral polarization while leaving the local discussion intact.

The central charge is a holomorphic section of the dual local system,

$$Z \in H^0(C^\circ, \Gamma_{\text{em}}^\vee \otimes_{\mathbb{Z}} \mathcal{O}_{C^\circ}),$$

meaning that $Z_\gamma(p)$ varies holomorphically with p for every locally constant charge γ . Define its components by

$$a^I := Z_{e_I}, \quad a_{D,I} := Z_{m^I}.$$

Then, for

$$\gamma = \sum_I (n_e^I e_I + n_{m,I} m^I), \quad n_e^I, n_{m,I} \in \mathbb{Z},$$

one has

$$Z_\gamma = \sum_I (n_e^I a^I + n_{m,I} a_{D,I}).$$

The integers $n_e^I, n_{m,I}$ are charges. The quantities $a^I, a_{D,I}$ are continuous complex central-charge values of basis charges and have mass dimension one; they are not integer charges. In a local electric Lagrangian frame, a^I is also the VEV of the complex scalar in the I -th Abelian vector multiplet. One can choose the local electric polarization so that

$$da^1, \dots, da^r$$

are linearly independent; the a^I are then local holomorphic *special coordinates* on C° .

This integral-charge/local-period structure is part of the physical consistency conditions on Coulomb geometry; see [9] for a systematic discussion in rank one.

3.6 Rigid special Kähler geometry

Proposition 3.11 (Local special geometry). *On C° , the two-derivative $\mathcal{N} = 2$ Abelian effective action makes the period map*

$$\Pi = \begin{pmatrix} a_{D,I} \\ a^I \end{pmatrix}$$

a holomorphic Lagrangian immersion into the local complex symplectic period space. Equivalently,

$$\sum_I da^I \wedge da_{D,I} = 0.$$

Consequently, on a sufficiently small electric coordinate patch there is a holomorphic prepotential $\mathcal{F}(a)$ such that

$$a_{D,I} = \frac{\partial \mathcal{F}}{\partial a^I},$$

and the effective gauge-coupling matrix is

$$\tau_{IJ} = \frac{\partial a_{D,I}}{\partial a^J} = \frac{\partial^2 \mathcal{F}}{\partial a^I \partial a^J}.$$

It is symmetric and satisfies $\text{Im } \tau > 0$ wherever the kinetic terms are positive and nondegenerate.

An intrinsic mathematical formulation of rigid special Kähler geometry and its relation to algebraic integrable systems is given by Freed [7].

The Lagrangian condition makes the holomorphic one-form

$$\vartheta = a_{D,I} da^I$$

closed. The local Poincaré lemma then gives $\vartheta = d\mathcal{F}$, explaining why the prepotential exists locally. The Kähler potential and metric can be written

$$K = \text{Im}(\overline{a^I} a_{D,I}), \quad g_{I\bar{J}} = \partial_I \partial_{\bar{J}} K = \text{Im} \tau_{IJ},$$

so

$$ds^2 = \text{Im} \tau_{IJ} da^I d\bar{a}^{\bar{J}}.$$

The same τ_{IJ} controls the Maxwell couplings, theta angles, and scalar kinetic terms.

This geometry is “special” because it is more than an ordinary Kähler manifold: it also carries the flat symplectic structure supplied by the electromagnetic local system. In a special frame,

$$x^I = \text{Re} a^I, \quad y_I = \text{Re} a_{D,I}$$

are local flat symplectic affine coordinates. Equivalently, the geometry can be described by a flat torsion-free symplectic connection compatible with the complex and Kähler structures.

Neither the symplectic basis nor the prepotential is usually global. Around a physical singular locus,

$$\Pi \longmapsto M\Pi, \quad M \in Sp(2r, \mathbb{Z})$$

in the principally polarized case, while the charge basis transforms contragrediently so that Z_γ is unchanged. The globally meaningful data are the charge local system, its pairing, the holomorphic central-charge section, and the induced metric.

The gauge-invariant chiral-ring coordinates

$$u_i = \langle \mathcal{O}_i \rangle$$

should not be confused with a^I . The u_i are intrinsic algebraic functions on C , whereas $a^I = a^I(u)$ are generally multivalued periods of dimension one. For example, in the semiclassical region of rank-one $SU(2)$ theory,

$$u = \langle \text{Tr} \phi^2 \rangle \sim a^2,$$

and the Weyl transformation $a \mapsto -a$ leaves u invariant.

3.7 Seiberg–Witten realization and degeneration

When an ordinary Seiberg–Witten curve description exists, it consists of a family of curves and a meromorphic differential, as in the original rank-one solutions [5, 6],

$$\pi : \Sigma \longrightarrow C, \quad \lambda_{\text{SW}}.$$

For $u \in C^\circ$, the fiber Σ_u is smooth. A locally chosen symplectic cycle basis $A_I, B^I \in H_1(\Sigma_u, \mathbb{Z})$ realizes the charge basis, and

$$a^I(u) = \oint_{A_I} \lambda_{\text{SW}}, \quad a_{D,I}(u) = \oint_{B^I} \lambda_{\text{SW}}, \quad Z_\gamma(u) = \oint_\gamma \lambda_{\text{SW}}.$$

For more general higher-rank theories, the appropriate object can be an algebraic integrable system rather than a single preferred curve; its periods encode the same special Kähler data.

At the physical discriminant, a cycle can degenerate and its period can tend to zero:

$$Z_\gamma(u) \longrightarrow 0.$$

When a BPS state of charge γ is present, its mass tends to zero and the Abelian effective action that had integrated it out ceases to be complete. The Seiberg–Witten family can often be extended over this locus with singular fibers, but the nonsingular rigid special Kähler effective geometry is defined only on C° . The conformal origin of an interacting SCFT is typically among these special degenerations.

Example 3.12 (Lagrangian gauge theory). In a Lagrangian $\mathcal{N} = 2$ gauge theory, the vector multiplet contains a complex adjoint scalar ϕ . A generic Coulomb vacuum has

$$\langle \phi \rangle \neq 0,$$

which can be placed in a Cartan subalgebra and breaks a rank- r gauge group G to its maximal torus T , locally

$$T \simeq U(1)^r.$$

For a root α , the corresponding non-Cartan W -boson has mass proportional to

$$|\alpha(\langle \phi \rangle)|.$$

This is nonzero at a generic point, so only the r Cartan photons remain massless. Gauge-invariant polynomials such as

$$u_k = \langle \text{Tr } \phi^k \rangle$$

provide Coulomb-branch coordinates.

The intrinsic definition above keeps the gauge-invariant operators $\text{Tr } \phi^k$ but does not require the existence of the elementary field ϕ .

Remark 3.13. The Coulomb chiral ring is often freely generated, but freeness is not part of the definition. If the generators obey relations, then C is a nontrivial affine variety rather than an affine space.

3.8 BPS particles and the occupied charge spectrum

From possible charges to actual particles

The charge lattice Γ_{em} , its Dirac pairing, and the central-charge section Z determine the allowed charge labels and the BPS mass bound in each sector. They do not determine whether a one-particle BPS state of a given charge actually exists. That additional information is the BPS spectrum.

It is useful to include flavor charges in a total charge lattice Γ_p . Locally over C° one may organize it as

$$0 \longrightarrow \Gamma_f \longrightarrow \Gamma_p \longrightarrow (\Gamma_{\text{em}})_p \longrightarrow 0.$$

The antisymmetric Dirac pairing extends to Γ_p and is degenerate along pure flavor charges:

$$\Gamma_f \subseteq \ker \langle \cdot, \cdot \rangle.$$

The central charge of

$$\gamma = (n_e^I, n_{m,I}, s_A) \in \Gamma_p$$

is the quantity already found in [section 3.4](#),

$$Z_\gamma(p) = n_e^I a^I(p) + n_{m,I} a_{D,I}(p) + s_A m^A.$$

Thus Γ_p lists possible superselection sectors and $Z_\gamma(p)$ fixes the lower mass bound in each sector. Neither statement asserts that the sector contains a state saturating that bound.

Definition 3.14 (Massive BPS particle). Fix $p \in C^\circ$. A one-particle state of charge $\gamma \in \Gamma_p$ is a *BPS particle* if its mass saturates the $\mathcal{N} = 2$ Poincaré bound,

$$M_\gamma(p) = |Z_\gamma(p)|.$$

It then belongs to a shortened massive representation of the Poincaré supersymmetry algebra.

The protected multiplicity is commonly recorded by the second helicity supertrace

$$\Omega(\gamma; p) = -\frac{1}{2} \text{Tr}_{\mathcal{H}_{\gamma,p}^{1p}} (-1)^{2J_3} (2J_3)^2,$$

where the trace is over the one-particle charge sector. In this convention a hypermultiplet contributes +1 and a vector multiplet contributes -2. Long multiplets cancel from the trace. A spin-refined protected character retains more information than Ω ; conversely, an accidental cancellation can make the unrefined index vanish even when BPS multiplets are present. Accordingly, Ω is a protected coarse-graining of the spectrum rather than a literal head count.

A lattice point is not a particle

$$\gamma \in \Gamma_p \not\Rightarrow \text{a BPS particle of charge } \gamma \text{ exists.}$$

The lattice specifies allowed charge sectors. Occupation of those sectors is dynamical information measured, at a protected level, by BPS indices.

These particles must also be distinguished from the protected local operators used to define $\mathcal{R}_{\text{CB}}$ and $\mathcal{R}_{\text{HB}}$. The latter are superconformal local-operator multiplets at the conformal vacuum. A massive BPS particle at $p \in C^\circ$ is instead an asymptotic excitation above a Poincaré-invariant Coulomb vacuum. It is classified by the massive Poincaré supersymmetry algebra and is not obtained by simply reinterpreting a Coulomb- or Higgs-branch chiral primary.

At the conformal origin, the periods of charges responsible for the degeneration can tend to zero. The corresponding excitations are then no longer separated finite-mass asymptotic particles, so an ordinary massive-particle spectrum need not be a valid description exactly at the origin. Consequently, the phrase “the BPS spectrum of an SCFT” normally refers to spectra in nearby generic Coulomb vacua, possibly after generic mass or Coulomb-branch deformations, together with their wall-crossing and degeneration data.

3.9 Walls of marginal stability and BPS chambers

For a nonzero central charge define the BPS phase and ray by

$$\phi_\gamma(p) = \arg Z_\gamma(p), \quad \ell_\gamma(p) = Z_\gamma(p) \mathbb{R}_{>0} \subset \mathbb{C}^*.$$

Suppose a state of charge $\gamma = \gamma_1 + \gamma_2$ can split into constituents of charges γ_1 and γ_2 . The triangle inequality gives

$$|Z_{\gamma_1+\gamma_2}| \leq |Z_{\gamma_1}| + |Z_{\gamma_2}|.$$

The decay reaches threshold precisely when the two constituent central charges have the same phase,

$$\frac{Z_{\gamma_1}(p)}{Z_{\gamma_2}(p)} \in \mathbb{R}_{>0}, \quad \text{equivalently} \quad \phi_{\gamma_1}(p) = \phi_{\gamma_2}(p),$$

so equality holds in the mass relation. This is a wall of marginal stability for that possible decay.

Definition 3.15 (BPS chamber). A *BPS chamber* is a connected region of the parameter space, away from the physical discriminant and all relevant marginal-stability walls, in which the protected BPS spectrum is constant. Within such a chamber, $\Omega(\gamma; p)$ is locally constant for every fixed locally transported charge γ .

Crossing a wall can create or destroy a bound state, and hence can change $\Omega(\gamma; p)$. The jumps are not arbitrary: the ordered product of BPS transformations remains invariant, which is the content of the Kontsevich–Soibelman wall-crossing formula. Its four-dimensional field-theory interpretation is developed in [14, 15].

Why a real wall appears inside a complex moduli space

The wall does not require a holomorphic period to vanish. It requires two nonzero complex numbers to align in phase, which is one real condition. Marginal-stability walls are therefore generally real codimension one and should not be confused with the complex-codimension-one physical discriminant where an additional charged state becomes massless.

There is therefore no vacuum-independent list of massive BPS particles attached directly to the conformal origin. A more invariant package consists of the spectra in the various chambers and the wall-crossing relations that connect them.

3.10 BPS quivers and stable representations

The BPS-quiver method gives a finite algebraic presentation of this particle sector when an additional positivity hypothesis holds. It is a construction internal to four-dimensional $\mathcal{N} = 2$ field theory: its inputs are a Coulomb vacuum, the charge lattice and pairing, central charges, and a choice of BPS chamber [12, 13].

Fix $p \in \mathcal{C}^\circ$ away from a marginal-stability wall and choose an open half-plane of width π ,

$$H_\vartheta = \{z \in \mathbb{C}^* \mid \vartheta < \arg z < \vartheta + \pi\},$$

whose boundary contains no BPS ray. By convention, BPS states with $Z_\gamma(p) \in H_\vartheta$ are called particles. Their CPT conjugates have charge $-\gamma$ and central charge $-Z_\gamma$, and lie in the opposite half-plane.

The half-plane fixes an orientation on charges of *actual* BPS states; it does not turn every lattice point with $Z_\gamma(p) \in H_\vartheta$ into a particle. In particular, one must not first take the entire lattice half-cone and choose arbitrary generators for it. The generators used below must themselves be occupied by stable elementary BPS states in the chosen chamber.

Definition 3.16 (Finite positive BPS basis). A *finite positive BPS basis* for (p, H_ϑ) is a finite set of charges

$$\gamma_1, \dots, \gamma_n$$

such that:

1. the γ_i form an integral basis of the relevant charge lattice;
2. each γ_i is represented by an actual stable BPS hypermultiplet with $Z_{\gamma_i}(p) \in H_\vartheta$;
3. every actual stable BPS particle charge in H_ϑ has a unique expansion

$$\gamma = \sum_{i=1}^n d_i \gamma_i, \quad d_i \in \mathbb{Z}_{\geq 0}.$$

Positivity does not manufacture the node particles

The definition is not an algorithm that starts only from (Γ_p, Z) , selects arbitrary lattice generators in H_ϑ , and then declares them BPS particles. Physical existence and stability of the elementary hypermultiplets of charges γ_i are inputs to the BPS-quiver presentation. The positive-basis condition then asserts their completeness as charge generators for the actual BPS particles in that half-plane.

Why mention all BPS particles if the immediate goal is only the quiver? It is not necessary to enumerate the complete spectrum in order to draw a candidate graph, but some independent argument must establish the completeness statement

$$Z_{\gamma_{\text{BPS}}}(p) \in H_\vartheta \implies \gamma_{\text{BPS}} = \sum_i d_i \gamma_i, \quad d_i \geq 0.$$

Without that implication one has a quiver built from some known BPS particles, not yet a complete BPS quiver for the chosen half-plane. A finite-length physical heart, discussed in [section 3.11](#), is one way to prove this implication without first listing every stable BPS state.

The quiver Q associated with such a basis is defined as follows:

- there is one node i for each stable basis hypermultiplet of charge γ_i ;
- its signed adjacency matrix is the Dirac pairing

$$B_{ij} = \langle \gamma_i, \gamma_j \rangle;$$

- with the arrow convention used here, $B_{ij} > 0$ gives B_{ij} arrows from node i to node j .

Reversing the convention for the pairing reverses all arrows. The invariant statement is that signed arrow multiplicities reproduce the antisymmetric Dirac pairing. In particular, electrically and magnetically nonlocal basis states are connected, while a pure flavor direction lies in the radical of B .

The graph is not the full datum. Oriented cycles can support a superpotential W , a cyclic polynomial in the arrows, and the central charges

$$Z_i := Z_{\gamma_i}(p)$$

determine stability. The appropriate object is therefore a quiver with potential and stability data,

$$(Q, W; Z_i),$$

not an unadorned directed graph. The pairing fixes B , but in general it does not determine W .

Why a representation appears only after the nodes are fixed

The quiver is a list of elementary constituents together with their interaction channels; it is not already the complete BPS spectrum. For a candidate total charge

$$\gamma = \sum_i d_i \gamma_i,$$

the integer d_i records how many constituents of type i are used. The space $V_i \simeq \mathbb{C}^{d_i}$ packages their multiplicity, while a matrix on an arrow records how constituents at its two ends are internally connected. A quiver representation is precisely this candidate bound-state configuration. Stability is then asked of the composite configuration, not used retroactively to prove that the physical node particles exist.

Now let

$$\gamma = \sum_i d_i \gamma_i.$$

A representation E of Q with dimension vector $\mathbf{d} = (d_1, \dots, d_n)$ assigns

$$V_i \simeq \mathbb{C}^{d_i}$$

to node i and a linear map

$$X_a : V_i \longrightarrow V_j$$

to every arrow $a : i \rightarrow j$. Its total charge and central charge are

$$\gamma(E) = \sum_i d_i \gamma_i, \quad Z(E) = \sum_i d_i Z_i.$$

The superpotential imposes the F-term relations

$$\frac{\partial W}{\partial X_a} = 0.$$

A *subrepresentation* $E' \subset E$ consists of subspaces $U_i \subseteq V_i$ preserved by every arrow map:

$$X_a(U_i) \subseteq U_j \quad \text{for every } a : i \longrightarrow j.$$

It represents a collection of constituents that is closed under all the connection maps and can therefore serve as a candidate decay subsystem. Its charge is

$$\gamma(E') = \sum_i (\dim U_i) \gamma_i.$$

Not every solution represents a one-particle bound state. It must also be stable. In one common phase convention, a nonzero representation E is Z -stable when every nonzero proper subrepresentation $E' \subset E$ obeys

$$\arg Z(E') < \arg Z(E).$$

Reversing the particle-half-plane or arrow convention reverses this displayed inequality without changing the physical content. At a marginal-stability wall the inequality becomes an equality and the representation can become strictly semistable.

Proposition 3.17 (Node simples are automatically stable). *For each node i , let S_i be the representation with*

$$(S_i)_i = \mathbb{C}, \quad (S_i)_j = 0 \quad (j \neq i),$$

and all arrow maps zero. Provided $Z_{\gamma_i} \in H_{\mathfrak{g}}$, the representation S_i is Z -stable.

Proof. The only subrepresentations of S_i are 0 and S_i itself. There is therefore no nonzero proper subrepresentation on which the stability inequality could fail. $\square$

This proposition has a deliberately limited logical role. Once a physical node hypermultiplet has been selected, its quiver avatar is a simple representation and hence is automatically stable inside the representation category. The proposition does *not* turn an arbitrary lattice charge into an occupied physical BPS sector. The statement

$$S_i \text{ is a stable simple representation}$$

and the statement

$$\gamma_i \text{ is carried by an actual elementary BPS hypermultiplet}$$

must not be silently identified; the latter is part of the physical construction of the quiver.

The moduli space relevant to charge γ is schematically

$$\mathcal{M}_d^{\text{st}}(Q, W; Z) = \left\{ (X_a) \left| \frac{\partial W}{\partial X_a} = 0, \quad Z\text{-stable} \right. \right\} / \prod_i GL(d_i, \mathbb{C}).$$

Under the BPS-quiver hypothesis, nonempty stable moduli spaces give the BPS bound states. Their cohomology, after the universal center-of-mass multiplet is treated appropriately, determines protected spin information; the corresponding Euler-characteristic specialization gives the unrefined index. Thus a node is a basic hypermultiplet, whereas more general hypermultiplets, vector multiplets, and higher-spin content can emerge from stable families of representations.

Example 3.18 (The two-node quiver). Let

$$\langle \gamma_1, \gamma_2 \rangle = 1, \quad Q : 1 \longrightarrow 2, \quad W = 0.$$

The simple representations at the two nodes are the quiver avatars of the already-chosen basis hypermultiplets of charges γ_1 and γ_2 . A representation of dimension vector $(1, 1)$ has one map $X : \mathbb{C} \rightarrow \mathbb{C}$. When $X = 0$, it splits as $S_1 \oplus S_2$ and describes two unbound constituents rather than a new one-particle state. When $X \neq 0$, it is a nontrivial candidate bound configuration; all such nonzero maps are equivalent under $GL(1, \mathbb{C})^2$, so the moduli space is a point.

With the stability convention above, its only nonzero proper subrepresentation has charge γ_2 . If the phases lie in the chosen half-plane and

$$\arg Z_{\gamma_2} < \arg Z_{\gamma_1},$$

then $\arg Z_{\gamma_2} < \arg Z_{\gamma_1 + \gamma_2}$ and the representation is stable. It gives a bound-state hypermultiplet of charge $\gamma_1 + \gamma_2$. At

$$\arg Z_{\gamma_1} = \arg Z_{\gamma_2}$$

the representation becomes marginal; on the other side it is unstable and that bound state is absent. This elementary example displays wall crossing as a change in stability rather than a change in the charge lattice.

What a BPS quiver does not describe

A BPS quiver is not the Coulomb chiral ring, the Higgs-branch coordinate ring, or a graph of vacua. It is a chamber- and basis-dependent presentation of massive BPS particles and their possible bound states above a chosen Coulomb vacuum.

3.11 A finite-length heart and the remaining physical gap

The positive-basis definition can appear circular. To prove that the γ_i generate every BPS particle charge, must one first know the complete BPS spectrum that the quiver was meant to help determine? Not necessarily. A categorical description supplies a sufficient noncircular mechanism, provided the category and stability condition have already been identified with the physical BPS sector.

Only a small amount of categorical language is needed here. Let $\mathcal{D}_{\text{BPS}}$ be a category proposed to describe the relevant particle sector, and let

$$(Z_{\text{phys}}, \mathcal{A}_H)$$

denote stability data consisting of the physical central charge and an Abelian heart $\mathcal{A}_H$ adapted to the particle half-plane H . The heart is the setting in which particle objects, subobjects, extensions, and stability inequalities are defined. The pairing of a central charge with such a heart is the basic structure formalized by Bridgeland stability, building on the physical idea of Π -stability [16, 17].

Definition 3.19 (Finite-length heart). An Abelian category $\mathcal{A}_H$ is *finite length* if every nonzero object E has a finite composition series

$$0 = E_0 \subset E_1 \subset \cdots \subset E_m = E$$

whose successive quotients E_k/E_{k-1} are simple objects. For the finite BPS-quiver mechanism, one additionally requires only finitely many simple isomorphism classes,

$$S_1, \dots, S_n.$$

Proposition 3.20 (What finite length actually proves). Assume that $\mathcal{A}_H$ is a finite-length heart with finitely many simples S_i , that $Z_{\text{phys}}(S_i) \in H$, and that stable objects of $\mathcal{A}_H$ describe all physical BPS particles whose central charges lie in H . Then:

1. every S_i is Z_{phys} -stable;
2. every object $E \in \mathcal{A}_H$ has

$$[E] = \sum_i d_i [S_i], \quad d_i \in \mathbb{Z}_{\geq 0};$$

3. in particular, the charge of every physical BPS particle in H is a nonnegative integral combination of the node charges

$$\gamma_i = [S_i].$$

Proof. A simple object has no nonzero proper subobject, so the stability condition is vacuous for each S_i . A composition series expresses every object as a finite succession of extensions of the simples. Additivity in the Grothendieck group gives the displayed charge decomposition, and the Jordan–Hölder theorem makes the multiplicities d_i independent of the chosen composition series. $\square$

This proposition proves generation of *charges* and construction of all objects by finite extensions. It does not say that the simples are the only stable objects, nor that every positive combination is stable. A nontrivial extension of simples can itself be stable and describe a composite BPS particle; this is exactly the sector tested by quiver representations.

Once the physical heart is established, its Ext-quiver has one node for each S_i . In the usual three-Calabi–Yau quiver setting, the antisymmetrized first-extension multiplicity reproduces the Dirac pairing,

$$\dim \text{Ext}^1(S_i, S_j) - \dim \text{Ext}^1(S_j, S_i) = \langle \gamma_i, \gamma_j \rangle,$$

up to the global arrow convention, while higher composition data determine the relations or superpotential W . This supplies the quiver from the heart without first enumerating all stable composite objects.

Remark 3.21 (Stability is not rigidity). Simplicity implies stability, but it does not by itself imply that a node is an isolated elementary hypermultiplet. A standard sufficient rigidity condition is

$$\text{Ext}^1(S_i, S_i) = 0.$$

If a simple object has self-deformations, quantizing its internal moduli can produce more complicated BPS spin content. Thus the usual node-hypermultiplet interpretation includes both the physical-state identification and the appropriate rigidity property.

A categorical quiver is not automatically the physical BPS quiver

An abstract or geometrically constructed category with a finite-length heart proves stability and positive generation only *inside that category*. To identify its Ext-quiver with the BPS quiver of the target four-dimensional theory, one must additionally justify

$$\begin{aligned} K_0(\mathcal{D}_{\text{BPS}}) &\simeq \Gamma_{\text{physical}}, & \chi(E, F) - \chi(F, E) &= \pm \langle \gamma_E, \gamma_F \rangle_{\text{DSZ}}, \\ Z_{\text{categorical}} &= Z_{\text{physical}}, & \mathcal{A}_H &= \text{the heart of an actual physical chamber,} \end{aligned}$$

and show that its stable objects quantize to finite-mass, normalizable one-particle BPS states. The sign in the pairing line is the global arrow convention. When a later geometric construction begins with a category such as a suitable compactly supported sector of $DCoh(X)$, these identifications are physical work to be done; they do not follow merely from the existence of simples and arrows.

The noncircular chain

$$\begin{aligned} &\text{physical BPS category} + Z_{\text{phys}} + \text{compatible finite-length heart } \mathcal{A}_H \\ &\quad \Downarrow \\ &\text{finitely many simples } S_i \quad \Longrightarrow \quad \text{stable node objects and positive charge generation} \\ &\quad \Downarrow \\ &\quad \text{Ext-quiver with potential } (Q, W). \end{aligned}$$

The first line is the substantive physical identification. The remaining implications are categorical.

3.12 Mutation and the scope of the quiver description

Keep the physical vacuum and spectrum fixed, but rotate the boundary of H_ϑ . When the ray of a basis particle γ_k leaves the half-plane, the ray of its antiparticle $-\gamma_k$ enters from the opposite side. The old positive basis is no longer adapted to the particle convention and must be replaced. For one orientation of the rotation, a convenient basis change is

$$\gamma'_k = -\gamma_k, \quad \gamma'_i = \gamma_i + [B_{ik}]_+ \gamma_k \quad (i \neq k), \quad [x]_+ := \max(x, 0).$$

Its pairing matrix is the quiver mutation at node k :

$$B'_{ij} = \begin{cases} -B_{ij}, & i = k \text{ or } j = k, \\ B_{ij} + [B_{ik}]_+ [B_{kj}]_+ - [-B_{ik}]_+ [-B_{kj}]_+, & i, j \neq k. \end{cases}$$

The opposite rotation gives the corresponding inverse convention. For a quiver with potential, W must be mutated as well and trivial two-cycles are removed; mutating only the visible graph can lose physical information. Mutation implements a change of positive basis and a dual description of the same BPS sector [13].

Wall crossing and mutation are therefore related but distinct operations:

	Wall crossing	Quiver mutation
What is varied	the vacuum or deformation parameters, so central-charge phases align	the particle half-plane or positive basis, possibly at fixed physical parameters
What changes	the set of stable BPS bound states and potentially $\Omega(\gamma; p)$	the quiver presentation $(Q, W; Z_i)$ of the same physical spectrum
Invariant content	constrained by the wall-crossing formula	recovered by identifying mutation-equivalent presentations

In a concrete path through parameter space the two changes can occur together, but they answer different questions: wall crossing is a physical change of stability, while mutation by itself is a change of description.

Accordingly, no single directed graph should be advertised as an absolute invariant of the SCFT. The more robust package is the mutation class of quivers with potential, together with the central-charge stability data and the wall-crossing relations among chambers.

Finally, the existence of a finite BPS quiver is an extra property, not a consequence of the $\mathcal{N} = 2$ algebra alone. The construction requires a half-plane and a finite positive basis of BPS hypermultiplets. Some theories or regions may fail this condition, for example because no such finite positive cone is available or because BPS rays have obstructive accumulation behavior. An infinite BPS spectrum does *not* by itself rule out a finite quiver: a finite quiver can have infinitely many stable representations. Thus the accurate scope statement is that a wide class of four-dimensional $\mathcal{N} = 2$ theories admits BPS-quiver descriptions in suitable chambers, not that every SCFT carries one canonical finite quiver.

4 The Higgs Branch

The Higgs-side dictionary

A highest-weight $\widehat{\mathcal{B}}_R$ primary gives a holomorphic Higgs-branch coordinate after one complex structure is chosen. The full real branch is a hyperkähler cone: $SU(2)_R$ rotates its complex structures, while flavor symmetry preserves each one and is generated by a triplet of moment maps.

4.1 Higgs-branch operators

Definition 4.1 ($\widehat{\mathcal{B}}_R$ primary and Higgs-branch chiral operator). Let $R \in \frac{1}{2}\mathbb{Z}_{\geq 0}$. A $\widehat{\mathcal{B}}_R$ superconformal primary is a Lorentz scalar with

$$j_1 = j_2 = 0, \quad r = 0, \quad \Delta = 2R,$$

transforming in the spin- R representation of $SU(2)_R$. In doublet-index notation it can be written

$$\mathcal{O}^{(i_1 \dots i_{2R})}, \quad i_a = 1, 2,$$

with all $2R$ indices symmetrized. In the convention where the index 1 has R_3 weight $+\frac{1}{2}$, its highest-weight component is

$$\mathcal{O}_{\text{HW}} := \mathcal{O}^{11 \dots 1}, \quad R_3 \mathcal{O}_{\text{HW}} = R \mathcal{O}_{\text{HW}}.$$

It obeys the shortening conditions

$$Q_\alpha^1 \mathcal{O}_{\text{HW}}(0) = 0, \quad \widetilde{Q}_{2\dot{\alpha}} \mathcal{O}_{\text{HW}}(0) = 0, \quad \alpha, \dot{\alpha} = 1, 2.$$

This highest-weight component is called a *Higgs-branch chiral operator*.

The $\widehat{\mathcal{B}}_R$ notation and null-state structure follow the superconformal multiplet classification [1]; the same protected multiplets play a central role in the four-dimensional chiral-algebra construction [2].

The $2R$ indices are a notation for the standard realization

$$\text{Sym}^{2R}(\mathbf{2})$$

of the spin- R irreducible representation of $SU(2)$. A completely symmetric tensor with $2R$ binary indices has $2R + 1$ independent components, with R_3 weights

$$R, R - 1, \dots, -R.$$

Thus $2R$ is the number of fundamental doublet indices, not the number of additional physical degrees of freedom. For example,

$$R = \frac{1}{2} : \mathcal{O}^i, \quad R = 1 : \mathcal{O}^{(ij)}, \quad R = \frac{3}{2} : \mathcal{O}^{(ijk)}.$$

The supercharge annihilation is an independent shortening condition, not a consequence of being highest weight alone. In radial quantization, let

$$|\mathcal{O}\rangle = \mathcal{O}_{\text{HW}}(0)|0\rangle.$$

Because $(Q_\alpha^I)^\dagger = S_I^\alpha$ and $(\tilde{Q}_{I\dot{\alpha}})^\dagger = \tilde{S}^{I\dot{\alpha}}$, the superconformal algebra gives, up to positive convention-dependent factors,

$$\|Q_\alpha^1|\mathcal{O}\rangle\|^2 \propto (\Delta - 2R - r)\|\mathcal{O}\|^2,$$

and

$$\|\tilde{Q}_{2\dot{\alpha}}|\mathcal{O}\rangle\|^2 \propto (\Delta - 2R + r)\|\mathcal{O}\|^2.$$

For $\Delta = 2R$ and $r = 0$, these descendants have zero norm and are set to zero in a unitary theory. By contrast, for $R > 0$ the complementary descendants $Q_\alpha^2|\mathcal{O}\rangle$ and $\tilde{Q}_{1\dot{\alpha}}|\mathcal{O}\rangle$ have nonzero norm in general. Hence the highest-weight operator is annihilated by four of the eight Poincaré supercharges and is 1/2-BPS. Raising or lowering $SU(2)_R$ indices may exchange the displayed labels 1 and 2 in other conventions.

What actually proves the shortening

$$\Delta = 2R, \quad r = 0 \implies \|Q^1\mathcal{O}_{\text{HW}}\|^2 = \|\tilde{Q}_2\mathcal{O}_{\text{HW}}\|^2 = 0 \implies Q^1\mathcal{O}_{\text{HW}} = \tilde{Q}_2\mathcal{O}_{\text{HW}} = 0.$$

The first implication uses the superconformal anticommutators; the second uses unitarity and quotienting by null states. Highest weight by itself is not enough.

The cases $\widehat{\mathcal{B}}_0$, $\widehat{\mathcal{B}}_{1/2}$, and $\widehat{\mathcal{B}}_1$ contain, respectively, the identity, a free-hypermultiplet scalar, and a flavor moment-map primary. Selecting the highest weight chooses an R_3 direction and hence one complex structure from the hyperkähler S^2 of complex structures; in that complex structure $\mathcal{O}_{\text{HW}}$ is holomorphic.

Definition 4.2 (Higgs-branch chiral ring). The *Higgs-branch chiral ring* $\mathcal{R}_{\text{HB}}$ is the commutative ring formed by highest-weight Higgs-branch operators, modulo their operator relations.

Definition 4.3 (Higgs branch). The *Higgs branch* is the family of $\mathcal{N} = 2$ Poincaré-supersymmetric vacua parameterized by VEVs of Higgs-branch chiral operators, with Coulomb-type order parameters set to zero. In a chosen complex structure,

$$\mathcal{H} = \text{Spec } \mathcal{R}_{\text{HB}}.$$

As for the Coulomb branch, an ordinary complex-valued Higgs vacuum corresponds to a character $\mathcal{R}_{\text{HB}} \rightarrow \mathbb{C}$ and hence to a maximal ideal. The full $\text{Spec } \mathcal{R}_{\text{HB}}$ additionally retains its scheme and local-ring structure.

4.2 Hyperkähler geometry

Definition 4.4 (Hyperkähler manifold). A *hyperkähler manifold* is a Riemannian manifold (M, g) equipped with three integrable complex structures I, J, K satisfying

$$I^2 = J^2 = K^2 = IJK = -1,$$

such that g is Kähler with respect to each complex structure.

The associated Kähler forms are

$$\omega_I(X, Y) = g(IX, Y), \quad \omega_J(X, Y) = g(JX, Y), \quad \omega_K(X, Y) = g(KX, Y).$$

There are not merely three isolated complex structures. For every unit vector $\vec{n} = (n_1, n_2, n_3) \in S^2$,

$$I_{\vec{n}} = n_1 I + n_2 J + n_3 K$$

also satisfies $I_{\vec{n}}^2 = -1$. Thus a hyperkähler manifold carries an $S^2 \simeq \mathbb{CP}^1$ family of complex structures.

Proposition 4.5. *The smooth locus of the Higgs branch is hyperkähler. Its real dimension is divisible by four:*

$$\dim_{\mathbb{R}} \mathcal{H} = 4d_H,$$

where d_H is called the quaternionic dimension:

$$d_H = \dim_{\mathbb{H}} \mathcal{H}.$$

The relation between extended supersymmetric sigma models, hyperkähler geometry, and hyperkähler quotients was developed systematically by Hitchin, Karlhede, Lindström, and Roček [8].

Definition 4.6 (Hyperkähler cone). A hyperkähler cone is a hyperkähler space whose smooth locus has a closed homothetic vector field χ satisfying

$$\nabla_X \chi = X, \quad \text{and hence} \quad \mathcal{L}_\chi g = 2g.$$

Locally its metric has cone form

$$g = dr^2 + r^2 g_Y, \quad \chi = r \frac{\partial}{\partial r}.$$

The vector fields

$$k_I = I\chi, \quad k_J = J\chi, \quad k_K = K\chi$$

are Killing and generate an $SU(2)$ action that rotates the three complex structures. The Higgs branch of a four-dimensional $\mathcal{N} = 2$ SCFT is such a cone, possibly singular at its tip and along lower-dimensional strata.

The radial homothety is the geometric action of dilatations on Higgs-branch VEVs. The rotating $SU(2)$ is the geometric realization of $SU(2)_R$: on the triple (I, J, K) its action factors through $SO(3) = SU(2)/\{\pm 1\}$. After choosing, say, I as the holomorphic complex structure, the subgroup $U(1) \subset SU(2)_R$ preserving I combines with positive real scaling to give the holomorphic $\mathbb{C}^*$ cone action.

Flavor symmetry behaves differently. It belongs to the full SCFT, not only to its Higgs branch, but its action on $\mathcal{H}$ is by tri-holomorphic isometries. If k_a is the vector field generated by a flavor generator, then

$$\mathcal{L}_{k_a} g = 0, \quad \mathcal{L}_{k_a} I = \mathcal{L}_{k_a} J = \mathcal{L}_{k_a} K = 0.$$

Thus $SU(2)_R$ rotates the complex structures, whereas flavor symmetry preserves each one separately.

4.3 The flavor-current multiplet and moment maps

Let F be a continuous flavor group. For each generator $T^a \in \mathfrak{f}$ there is a conserved current

$$j_\mu^a, \quad \partial^\mu j_\mu^a = 0, \quad F^a = \int d^3x j_0^a.$$

In a four-dimensional CFT, a conserved vector current has

$$\Delta(j_\mu^a) = 3, \quad (j_1, j_2) = \left(\frac{1}{2}, \frac{1}{2}\right).$$

Because a flavor charge commutes with the full superconformal algebra, this current is an $SU(2)_R \times U(1)_r$ singlet.

The current is not the superconformal primary of its multiplet. Conservation, shortening, and the condition $[F^a, Q] = 0$ place an ordinary flavor current in a unique $\mathcal{N} = 2$ current multiplet [1], conventionally denoted

$$\widehat{\mathcal{B}}_1.$$

Its superconformal primary is a Lorentz scalar $\mu^{a(ij)} = \mu^{a(ji)}$ with

$$\Delta = 2, \quad R = 1, \quad r = 0.$$

The current occurs at the $Q\widetilde{Q}$ descendant level, schematically

$$j_{\alpha\dot{\alpha}}^a \sim Q_\alpha^i \widetilde{Q}_{\dot{\alpha}}^j \mu_{ij}^a + \text{terms selecting the conserved conformal primary.}$$

The dimension count $2 + \frac{1}{2} + \frac{1}{2} = 3$ and the Lorentz product $(\frac{1}{2}, 0) \otimes (0, \frac{1}{2})$ explain how the vector can occur, but they do not by themselves prove uniqueness. The uniqueness statement uses the classification of shortened superconformal multiplets containing conserved currents. In another common notation the same multiplet is

$$B_1 \overline{B}_1[0; 0]_2^{(2;0)},$$

where the $SU(2)_R$ Dynkin label 2 means spin $R = 1$.

By contrast, the stress-tensor multiplet contains the $SU(2)_R \times U(1)_r$ currents. Their charges rotate the supercharges and therefore are not flavor charges.

The moment-map primary and its protected correlators are also reviewed from the chiral-algebra viewpoint in [2].

Definition 4.7 (Flavor moment-map operator). The primary

$$\mu^{a(ij)} \in \text{adj}(F) \otimes \mathfrak{z}_{SU(2)_R}$$

of the $\widehat{\mathcal{B}}_1$ flavor-current multiplet is called the *flavor moment-map operator*. Its highest-weight component, for example $\mu^{a(11)}$, is a Higgs-branch chiral operator.

The terminology has a direct geometric meaning. Let k_a generate the flavor action on the smooth Higgs branch. Since this action preserves all three Kähler forms, it has a triplet of moment maps satisfying, up to an overall sign convention,

$$d\mu_I^a = \iota_{k_a} \omega_I, \quad d\mu_J^a = \iota_{k_a} \omega_J, \quad d\mu_K^a = \iota_{k_a} \omega_K.$$

These three real functions are the geometric realization of the three components of $\mu^{a(ij)}$. In complex structure I , one combines

$$\Omega_I = \omega_J + i\omega_K, \quad \mu_{\mathbb{C}}^a = \mu_J^a + i\mu_K^a,$$

so that

$$d\mu_{\mathbb{C}}^a = \iota_{k_a} \Omega_I.$$

The holomorphic function $\mu_{\mathbb{C}}^a$ is the chosen highest-weight component in the Higgs chiral ring.

Higgs-branch operators may transform in nontrivial flavor representations, so flavor transformations can move points of $\mathcal{H}$ and a Higgs VEV may break F to the stabilizer of the chosen point. This is why flavor moment maps naturally appear in the Higgs-branch discussion. Flavor symmetry still belongs to the whole SCFT: it acts on BPS states above the Coulomb branch and its background masses deform the Coulomb special geometry, even though continuous flavor acts trivially on the pure Coulomb base itself.

Example 4.8 (Lagrangian hyperkähler quotient). In a Lagrangian gauge theory with hypermultiplet scalar space $\mathcal{X}$ and gauge group G , the classical Higgs branch is

$$\mathcal{H} = \mu_G^{-1}(0)/G,$$

where μ_G is the triplet of *gauge* moment maps. These are constraints used to remove gauge redundancy: one imposes $\mu_G = 0$ and divides by G . Equivalently, in a chosen complex structure one imposes the complex gauge-moment-map equation and takes the quotient by $G_{\mathbb{C}}$, with the appropriate stability condition.

This must be distinguished from a flavor moment map μ_F . A surviving flavor group is a genuine global symmetry of the quotient, and μ_F descends to a gauge-invariant physical function on $\mathcal{H}$. Thus gauge moment maps construct the Higgs branch, while flavor moment maps are operators living on it. The intrinsic SCFT definition replaces elementary hypermultiplet fields by these gauge-invariant operators and their ring relations.

5 Coulomb, Higgs, and Mixed Branches

	Coulomb branch	Higgs branch	Mixed branch
Order parameters	Coulomb chiral operators	$\widehat{\mathcal{B}}_R$ operators	Both types
$SU(2)_R$ quantum number	$R = 0$	generally $R > 0$	both sectors present
$U(1)_r$ quantum number	nonzero, convention-dependent	$r = 0$	both sectors present
Natural geometry	rigid special Kähler on $\mathcal{C}^\circ$	hyperkähler cone	stratified combination of Coulomb and Higgs directions
Generic massless fields	r Abelian vector multiplets; neutral hypers may coexist	hypermultiplet moduli, possibly with a residual sector	vector and hypermultiplet moduli

Definition 5.1 (Mixed branch). A *mixed branch* is a family of supersymmetric vacua in which compatible Coulomb-branch and Higgs-branch operators simultaneously acquire nonzero VEVs.

Definition 5.2 (Enhanced Coulomb branch). An *enhanced Coulomb branch* is a mixed branch whose generic point lies over the Coulomb base and contains additional neutral massless hypermultiplet

directions. Locally over a regular patch it has the form

$$C^\circ \times \mathbb{H}^h$$

or, more generally, a hyperkähler fibration over C° . The integer h counts neutral hypermultiplets and does not change the Coulomb rank r .

Enhanced Coulomb branches and their role in rank-one SCFT geometry are analyzed in [10].

A mixed branch is usually stratified. Near a sufficiently generic smooth point of a stratum, its massless directions often locally separate into Coulomb-type and Higgs-type factors, although global product decompositions need not exist.

Remark 5.3. Statements such as “the Coulomb branch is given by $q = 0$ ” and “the Higgs branch is given by $\phi = 0$ ” belong to a particular Lagrangian presentation. The intrinsic distinction is instead determined by the superconformal multiplets of the operators acquiring VEVs.

6 Expanding Around a Vacuum

The infrared question

Fixing a vacuum chooses a superselection sector and generates mass thresholds. One first constructs a finite-energy EFT and only then takes its strict infrared limit. The tangent multiplets become free at a smooth point, but the complete answer may include an additional residual factor.

6.1 What an expansion around a vacuum means

Choose a vacuum $p \in \mathcal{M}_{\text{vac}}$ and a local operator with VEV

$$\langle O \rangle_p = v.$$

One writes schematically

$$O(x) = v + \delta O(x),$$

where δO represents fluctuations around the chosen vacuum.

Definition 6.1 (Low-energy effective field theory). A *low-energy effective field theory* is the theory governing degrees of freedom whose characteristic energies satisfy

$$E \ll M,$$

where M is the mass scale of the degrees of freedom that have been removed from the description.

Definition 6.2 (Integrating out). To *integrate out* a massive degree of freedom means to eliminate it from the explicit low-energy description while retaining its effects through effective interactions among the remaining light fields.

For a VEV v of dimension Δ , the natural separation scale is

$$M_v \sim |v|^{1/\Delta}.$$

At energies $E \ll M_v$, massive states created by the symmetry breaking may be integrated out.

Definition 6.3 (Massless degree of freedom). A *massless degree of freedom* is an excitation whose energy can become arbitrarily small as its spatial momentum tends to zero. Such fields remain in the deep infrared effective theory.

6.2 The conformal vacuum

At the tip of every branch,

$$\langle O_i \rangle = 0$$

for all positive-dimension order parameters. Expanding around this vacuum gives the original SCFT itself. This is the unique point at which the full superconformal symmetry is manifest in the vacuum.

Proposition 6.4. *The original interacting SCFT is recovered at the conformal origin, not at a generic nonzero-VEV vacuum.*

6.3 Finite-energy EFT versus the strict IR fixed point

A VEV is not a coupling deformation

Vacuum selection keeps the microscopic theory $\mathcal{T}$ fixed and chooses a state $|\Omega_p\rangle$ in which observables are evaluated. A coupling deformation instead changes the action, schematically

$$S \mapsto S + \lambda \int d^4x O(x).$$

Thus a VEV means “same theory, different state”, whereas changing λ moves in theory space. These operations can lead to similar-looking scales but are conceptually different.

In infinite volume, different vacua are superselection sectors selected by boundary conditions at spatial infinity. Wilsonian coarse graining is then performed inside the chosen sector, so the infrared endpoint naturally depends on the pair

$$(\mathcal{T}, p) \mapsto \mathcal{T}_{\text{IR}, p}.$$

Different vacua of the same microscopic theory can consequently have different infrared fixed points without the VEV becoming a coupling.

A nonzero VEV nevertheless generates a scale

$$M_p \sim |v|^{1/\Delta},$$

or, on a Coulomb branch, the collection of charged thresholds $|Z_\gamma(p)|$. At energies $E \ll M_p$, massive states can be integrated out and one obtains a Wilsonian effective action

$$S_{\text{EFT}, p} = S_2 + \sum_{d>4} \frac{c_d}{M_p^{d-4}} \int d^4x O_d.$$

At finite nonzero E/M_p , this action still remembers the VEV scale and is generally not conformal.

The *strict IR fixed point* is obtained by taking

$$E/M_p \longrightarrow 0.$$

Irrelevant interactions disappear in this limit. The resulting fixed point can be a free SCFT, an interacting SCFT, or a product of free and interacting factors. It need not equal the original SCFT at the conformal vacuum.

There is no contradiction between spontaneous breaking of the original dilatation symmetry and emergent conformal symmetry of the fluctuation theory. If

$$a(x) = a_0 + \delta a(x),$$

the original dilatation acts on δa with an inhomogeneous term proportional to a_0 , so the chosen vacuum is not invariant. The Gaussian strict-IR action can instead possess an emergent dilatation acting homogeneously on δa .

Finally, a massless degree of freedom need not be a modulus. The tangent space to a branch counts scalar fluctuations whose constant values move the vacuum, but an additional gapless or interacting sector can have no such vacuum direction. Consequently, proving that the tangent-mode sector is free does not prove that the complete IR theory contains nothing else.

The order of operations

choose $p \implies$ identify the mass scale $M_p \implies$ integrate out states with $M \gtrsim M_p$

$$\begin{array}{c} \Downarrow \\ S_{\text{EFT},p} \xrightarrow{E/M_p \rightarrow 0} \mathcal{T}_{\text{IR},p}. \end{array}$$

The finite-energy EFT generally remembers M_p and is not conformal. The strict IR fixed point is the limiting theory after irrelevant interactions have disappeared.

6.4 A generic Coulomb-branch vacuum

Let $p \in C^\circ$ be a generic physical regular point of a rank- r Coulomb branch. The low-energy massless fields contain r Abelian $\mathcal{N} = 2$ vector multiplets. In a chosen electric frame the gauge group is locally written

$$U(1)^r.$$

More invariantly, the local gauge algebra is $\mathfrak{u}(1)^r$; discrete quotients and the spectrum of genuine line operators are additional global data.

The special Kähler action describes the vector-moduli sector, not automatically the complete massless theory. Denote this part by S_{vec} . In local special coordinates, its bosonic terms include schematically

$$S_{\text{vec}} \supset \int d^4x \, \text{Im } \tau_{IJ}(a) \left(F_{\mu\nu}^I F^{I\mu\nu} + \partial_\mu a^I \partial^\mu \bar{a}^I \right) + \dots.$$

Expanding around $a^I = a_0^I$ gives

$$\tau_{IJ}(a) = \tau_{IJ}(a_0) + \partial_K \tau_{IJ}(a_0) \delta a^K + \dots.$$

The quadratic term describes free Maxwell multiplets with constant couplings. The first interaction contains terms such as

$$\partial_K \tau_{IJ}(a_0) \delta a^K F_{\mu\nu}^I F^{J\mu\nu}.$$

Since $[\delta a] = 1$ and $[F] = 2$, this is a dimension-five interaction whose coefficient has dimension -1 and is suppressed by the VEV scale. Higher powers and higher-derivative terms are likewise irrelevant. Therefore

$$(S_{\text{vec}})_{\text{strict IR}} = \{r \text{ free } \mathcal{N} = 2 \text{ Maxwell multiplets}\}.$$

This conclusion concerns only the vector-moduli sector. The complete low-energy action may have the schematic form

$$S_{\text{IR},p} = S_{\text{vec}} + S_{\text{neutral}} + S_{\text{mix}}.$$

If the mixing interactions are irrelevant, the strict IR fixed point factorizes as

$$\mathcal{T}_{\text{IR},p} \simeq \{r \text{ free Abelian vector multiplets}\} \times \mathcal{T}_{\text{neutral},p}.$$

Physical regularity excludes additional massless states carrying the electromagnetic charges of the $U(1)^r$ sector, but it does not exclude neutral massless fields. Moreover, an additional massless vector modulus would increase $\dim_{\mathbb{C}} C$ and is therefore already counted in r . The residual neutral factor can contain free neutral hypermultiplets, as on an enhanced Coulomb branch, or a decoupled rank-zero gapless sector. A genuinely interacting residual factor at a generic point is possible in principle but is more special than the usual free-vector situation.

Example 6.5 ($\mathcal{N} = 4$ Yang–Mills viewed as an $\mathcal{N} = 2$ theory). An $\mathcal{N} = 4$ vector multiplet decomposes into an $\mathcal{N} = 2$ vector multiplet and an adjoint hypermultiplet. At a generic rank- r Coulomb VEV, the non-Cartan fields are massive, while the Cartan fields give

$$r \text{ free } \mathcal{N} = 2 \text{ vectors} \quad + \quad r \text{ free neutral hypermultiplets}.$$

Equivalently, the strict IR contains r free Abelian $\mathcal{N} = 4$ multiplets. This is a standard example in which the Coulomb vector sector is not the whole massless theory.

6.5 A generic Higgs-branch vacuum

Let p be a smooth point of the Higgs branch. The scalar fluctuations tangent to $\mathcal{H}$ form hypermultiplets. Their two-derivative action is a nonlinear sigma model; to emphasize that this is only the moduli sector, write

$$S_{\text{moduli}} = \int d^4x \, g_{A\bar{B}}(X) \partial_\mu X^A \partial^\mu \bar{X}^{\bar{B}} + \dots.$$

Choose normal coordinates near p and write

$$X^A = X_0^A + \delta X^A.$$

Then

$$g_{A\bar{B}}(X) = g_{A\bar{B}}(X_0) + O(\delta X^2).$$

The leading term describes free hypermultiplet fluctuations associated with $T_p \mathcal{H}$. The first curvature correction is schematically

$$R_{A\bar{B}C\bar{D}}(X_0) \delta X^C \delta \bar{X}^{\bar{D}} \partial_\mu \delta X^A \partial^\mu \delta \bar{X}^{\bar{B}}.$$

It has dimension six in four spacetime dimensions and is irrelevant. Hence the tangent hypermultiplets become free in the strict IR.

Again, this proves a statement about the moduli sector rather than the complete IR theory. In general,

$$S_{\text{IR},p} = S_{\text{moduli}} + S_{\text{residual}} + S_{\text{mix}},$$

and, when the mixing flows to zero,

$$\mathcal{T}_{\text{IR},p} \simeq \{\text{free hypermultiplets from } T_p \mathcal{H}\} \times \mathcal{T}_{\text{residual},p}.$$

Smoothness of $\mathcal{H}$ only controls the vacuum directions parameterized by the Higgs chiral ring. It cannot exclude a massless sector with no Higgs modulus.

For example, if

$$\mathcal{T} = \mathcal{T}_1 \times \mathcal{T}_2,$$

where only $\mathcal{T}_1$ has a Higgs branch, then a VEV in $\mathcal{T}_1$ leaves $\mathcal{T}_2$ untouched. The infrared theory can therefore contain free tangent hypers times the residual theory $\mathcal{T}_2$, even at a smooth point of $\mathcal{H}$.

Residual vector multiplets require more care. At a generic point of a pure Higgs branch of an indecomposable theory, one normally expects no unbroken dynamical vector sector. If a gauge group is only partially Higgsed and massless vectors remain, their scalars generally supply Coulomb directions; this indicates a mixed branch or a boundary between Higgs and mixed strata. A decoupled vector factor is another possibility. Thus residual interacting sectors are compatible with a smooth Higgs branch, but residual vectors should not be asserted generically without this qualification.

If no residual sector is present, the complete strict-IR theory at a generic smooth Higgs point consists only of free hypermultiplets.

6.6 Singular vacua

Definition 6.6 (Geometric and physical singularity). A point $p \in \mathcal{M}_{\text{vac}}$ is *geometrically singular* if its local algebraic or differential moduli-space geometry fails to be smooth. A Coulomb vacuum is *physically singular* if it lies in Δ_{phys} , so the Abelian vector-multiplet effective description is singular or incomplete because additional charged degrees of freedom become massless.

The two notions are related but not identical. A geometric singularity often signals extra massless degrees of freedom, but a physically singular Coulomb point can lie on an algebraically smooth base. At the Coulomb discriminant, a charge γ can obey

$$Z_\gamma(p) = 0.$$

If a BPS state of this charge is present, it becomes massless and cannot be integrated out. A single mutually local collection can admit another weakly coupled description, while simultaneous mutually nonlocal massless states can produce an interacting fixed point of Argyres–Douglas type [11]. At a singular Higgs-branch locus, additional interacting directions or unbroken sectors may likewise appear.

The infrared theory at such a point can be

$$\mathcal{T}_{\text{IR},p} = \mathcal{T}'_{\text{SCFT}} \times \mathcal{T}_{\text{free}},$$

where $\mathcal{T}'_{\text{SCFT}}$ may be a new interacting SCFT and $\mathcal{T}_{\text{free}}$ contains decoupled free vector or hypermultiplets.

Remark 6.7. An interacting SCFT appearing at a singular point need not be identical to the original SCFT at the common origin. It may have lower rank, different flavor symmetry, different Coulomb dimensions, or different anomaly coefficients. The microscopic theory is unchanged by selecting p , but the strict IR fixed point of the sector $(\mathcal{T}, p)$ can be different because massive thresholds and the set of remaining massless states depend on the vacuum.

6.7 Summary by vacuum type

Vacuum	Preserved symmetry	Theory of low-energy fluctuations
Conformal origin	full $\mathcal{N} = 2$ superconformal symmetry	the original interacting SCFT
Generic physical regular Coulomb vacuum	$\mathcal{N} = 2$ Poincaré supersymmetry	r free Abelian vector multiplets, possibly times a neutral residual factor
Generic smooth Higgs vacuum	$\mathcal{N} = 2$ Poincaré supersymmetry	free tangent hypermultiplets, possibly times a residual factor; vectors indicate mixed or decoupled sectors
Generic smooth mixed vacuum	$\mathcal{N} = 2$ Poincaré supersymmetry	vector and hypermultiplet moduli, possibly times a residual sector
Singular vacuum	at least $\mathcal{N} = 2$ Poincaré supersymmetry	extra massless states; possibly a new interacting SCFT times free fields

7 Conceptual Dictionary

SCFT

A quantum field theory invariant under both conformal transformations and supersymmetry.

Local operator

An operator inserted at a spacetime point. Its products and correlation functions encode local measurements.

Operator spectrum

The collection of local operators organized by scaling dimension, Lorentz spin, R-symmetry, flavor symmetry, and superconformal multiplet.

OPE

The short-distance expansion of a product of local operators into a sum of local operators.

Cluster decomposition

Factorization of correlators at infinite spacelike separation in a pure vacuum. Together with protected position independence, it makes the VEV map multiplicative.

VEV

The expectation value of an operator in a chosen vacuum.

Vacuum

A translationally invariant state of minimum energy. A supersymmetric vacuum is additionally annihilated by the Poincaré supercharges.

Modulus

A continuous parameter labeling inequivalent vacua.

Massless degree of freedom versus modulus

A modulus becomes a massless scalar fluctuation, but the converse need not hold. A gapless sector can have an isolated vacuum and therefore contribute no tangent direction to the vacuum moduli space.

Vacuum moduli space

The space formed by all continuously connected supersymmetric vacua.

Branch

A distinguished component or stratum of the vacuum moduli space, classified by which protected operators acquire VEVs.

Chiral ring

A protected commutative ring of specified scalar primaries, with multiplication defined by the protected OPE. Its algebraic relations become equations for a moduli-space branch.

 $\widehat{\mathcal{B}}_R$ multiplet

A half-BPS multiplet whose scalar superconformal primary has $\Delta = 2R$, $r = 0$, and $SU(2)_R$ spin R . It may be written as a tensor with $2R$ symmetric doublet indices; its highest-weight component is a Higgs-branch chiral operator.

Flavor-current multiplet

The $\widehat{\mathcal{B}}_1$ multiplet associated with a continuous flavor generator. Its scalar primary $\mu^{a(ij)}$ is an $SU(2)_R$ triplet, while the conserved current j_μ^a is a superconformal descendant.

Flavor moment map

The $\widehat{\mathcal{B}}_1$ primary viewed as a triplet of Hamiltonians for the tri-holomorphic flavor action on the Higgs branch. It is a physical operator, unlike a gauge moment map used as a hyperkähler-quotient constraint.

Character and closed point

A unital $\mathbb{C}$ -algebra map $\chi_p : R \rightarrow \mathbb{C}$. Its kernel is the maximal ideal corresponding to the ordinary complex vacuum p .

Affine spectrum

$\text{Spec } R$ is the scheme of all prime ideals together with its local rings and structure sheaf. Its non-closed generic points are not additional ordinary complex vacua.

Coulomb branch

The branch parameterized by VEVs of $SU(2)_R$ -singlet Coulomb chiral primaries. Its physical regular locus is rigid special Kähler.

Higgs branch

The branch parameterized by VEVs of highest-weight $\widehat{\mathcal{B}}_R$ operators. It is a hyperkähler cone: dilatations generate the homothety, $SU(2)_R$ rotates its sphere of complex structures, and flavor symmetry preserves each complex structure.

Mixed branch

A branch carrying compatible VEVs of both Coulomb and Higgs types.

Enhanced Coulomb branch

A mixed branch with neutral hypermultiplet directions fibered over a generic Coulomb base.

Electromagnetic charge lattice

The integral local system $\Gamma_{\text{em}} \rightarrow C^\circ$ of electric and magnetic charges, equipped with the Dirac pairing and electric–magnetic monodromy.

Supersymmetry central charge

The center operator $\widehat{Z}$ in the $4d \mathcal{N} = 2$ Poincaré superalgebra. In a fixed vacuum it is a surface charge whose eigenvalue is determined by the full electric, magnetic, and flavor charge sector.

Special coordinates

The continuous complex period functions $a^I = Z_{eI}$ and $a_{D,I} = Z_{mI}$ associated with a local symplectic charge basis. They are not integer charges, and only the a^I are chosen as local coordinates in an electric frame.

Rigid special Kähler geometry

The Kähler metric together with the flat integral symplectic structure and holomorphic Lagrangian period map supplied by (Γ_{em}, Z) . Locally it is encoded by a prepotential $\mathcal{F}$.

Seiberg–Witten geometry

A curve or algebraic-integrable-system realization of the charge local system and central-charge periods. Its smooth fibers describe C° and its degenerations encode the physical discriminant.

Massive BPS particle

A one-particle excitation above a fixed Coulomb vacuum that saturates $M_\gamma = |Z_\gamma|$. It is a shortened representation of the $\mathcal{N} = 2$ Poincaré algebra, not a superconformal local-operator primary at the conformal vacuum.

BPS index

A protected helicity supertrace $\Omega(\gamma; p)$ of the one-particle charge sector. It is locally constant inside a BPS chamber and can jump across a wall of marginal stability.

BPS chamber

A connected region away from the physical discriminant and marginal-stability walls in which the protected BPS spectrum is constant. A marginal-stability wall occurs when central charges of possible decay constituents align in phase.

Finite positive BPS basis

A finite integral basis $\{\gamma_i\}$ whose members are charges of actual stable elementary BPS hypermultiplets in the chosen particle half-plane and which positively generates the charges of every actual BPS particle in that half-plane. It is not an arbitrary basis of lattice points selected by $Z_\gamma \in H$.

BPS quiver

A finite, chamber-dependent presentation of the massive BPS sector, when a finite positive basis exists. Nodes are already-stable physical basis hypermultiplets, signed arrows encode the Dirac pairing, W imposes relations, and Z supplies stability for composite representations.

Simple node representation

The representation S_i supported by a one-dimensional space at node i and zero spaces elsewhere. It is automatically stable because it has no nonzero proper subrepresentation. This categorical fact does not prove that an arbitrary charge γ_i is occupied by a physical BPS particle.

Finite-length heart

An Abelian heart in which every object has a finite composition series. If it is the physically correct heart and has finitely many simples, those simples are stable and their classes positively generate all particle-object charges. Finite length inside an abstract category does not by itself establish the physical identification.

Quiver mutation

A change of positive BPS basis induced, in its simplest use, by rotating the particle half-plane. It changes the quiver presentation without by itself changing the physical spectrum, and must not be identified with wall crossing.

Effective field theory

A description valid below a chosen energy scale, containing only the degrees of freedom that remain light at that scale.

Infrared limit

The limit of physical processes at energies much smaller than all massive scales.

Strict IR fixed point

The limiting scale-invariant theory obtained after all irrelevant interactions of a finite-energy EFT have flowed away. It can be different from the original SCFT even when the scale arose by choosing a vacuum rather than by deforming the action.

Residual sector

Massless degrees of freedom not parameterized by the tangent directions of the branch under discussion. At a physical regular Coulomb point such a sector must be electromagnetically neutral; on a pure Higgs branch residual vectors usually signal a mixed or decoupled factor.

Free field theory

A theory whose leading action is quadratic and whose asymptotic excitations do not interact.

Interacting fixed point

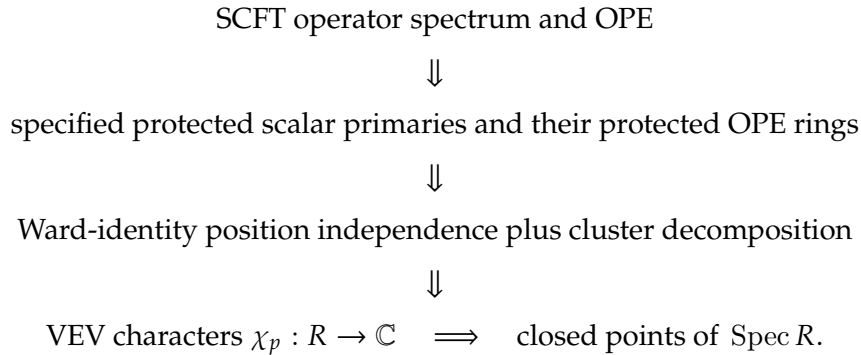
A scale-invariant infrared or ultraviolet limit with nontrivial OPE coefficients and genuine interactions.

Singular vacuum

A geometrically singular point or a physically singular point where an effective description degenerates. The notions often coincide but need not do so.

8 Master Reasoning Map

The intrinsic construction can be summarized as



The algebraic and low-energy constructions are distinct layers:

$\mathcal{R}_{\text{CB}} \implies C = \text{Spec } \mathcal{R}_{\text{CB}}$	complex algebraic or scheme data,
$p \in C^\circ \text{ and low-energy } \mathcal{N} = 2 \implies \mathfrak{u}(1)^r, \Gamma_{\text{em}}, Z$	dynamical electromagnetic data,
$(\Gamma_{\text{em}}, Z) \implies (a^I, a_{D,I}), \mathcal{F}, \tau, g$	rigid special Kähler data.

In particular, the photons and Seiberg–Witten periods do not follow from $\text{Spec } \mathcal{R}_{\text{CB}}$ alone.

The protected massive-particle sector requires another dynamical step, and the logical gap should remain visible:

$$\begin{array}{c}
 p \in C^\circ, \quad \Gamma_p, \quad \langle, \rangle, \quad Z \\
 \not\Rightarrow \\
 \text{occupied BPS sectors or node particles} \\
 \text{actual stable elementary hypermultiplets } \{\gamma_i\} + \text{positive-basis completeness} \\
 \Downarrow \\
 (Q, W; Z_i) \implies Z\text{-stable representations} \implies \text{BPS particles and bound states.}
 \end{array}$$

The first line cannot be shortened to (Γ_p, Z) alone: possible charge labels and their mass bounds do not determine which sectors are occupied. The nodes are stable physical particles before they are represented by quiver simples; simplicity explains their categorical stability but not their physical existence.

When a categorical model is available, completeness can instead be established without enumerating the spectrum:

$$\begin{array}{c}
 \text{physically identified } (\mathcal{D}_{\text{BPS}}, Z_{\text{phys}}, \mathcal{A}_H) + \text{finite length and finitely many simples} \\
 \Downarrow \\
 \{S_i\}_{\text{simple}} \implies \text{stable nodes and positive generation of all particle-object charges} \\
 \Downarrow \\
 \text{Ext-quiver with potential } (Q, W).
 \end{array}$$

The qualifier *physically identified* carries the remaining gap: an abstract or geometric heart becomes the target four-dimensional BPS heart only after its charges, pairing, central charge, chamber, and stable one-particle objects have been matched to the physical theory.

After choosing a vacuum p , one studies fluctuations by

$$\text{SCFT} \xrightarrow{\text{choose a BPS VEV}} \mathcal{N} = 2 \text{ Poincaré vacuum} \xrightarrow{E \ll M_v} \text{infrared effective theory near } p.$$

Thus the origin gives the original SCFT; a regular point gives free moduli fields, possibly with a residual sector; and a singular point can add massless states or a new interacting SCFT.

More precisely, branch geometry determines the tangent multiplets but does not by itself exhaust the massless spectrum:

$$\mathcal{T}_{\text{IR}, p} \simeq \mathcal{T}_{\text{tangent, free}, p} \times \mathcal{T}_{\text{residual}, p}.$$

For a physical regular Coulomb vacuum, the tangent factor consists of r free Abelian vector multiplets and any residual factor is electromagnetically neutral. For a smooth Higgs vacuum, the tangent factor consists of free hypermultiplets; residual vectors require a mixed-branch or decoupled-sector qualification.

Minimal set of statements to remember

1. A branch coordinate is a specified protected scalar primary, not an entire multiplet.
2. A pure vacuum makes VEVs multiplicative only after protected position independence and cluster decomposition are used.
3. Ordinary complex vacua are characters, equivalently maximal ideals under finite generation; Spec contains additional geometric bookkeeping.
4. Coulomb geometry as an affine variety is not yet Seiberg–Witten geometry. The latter needs the low-energy charge lattice and central-charge section.
5. a^I and $a_{D,I}$ are continuous period values; n_e^I and $n_{m,I}$ are the integer charges.
6. The charge lattice lists possible sectors; it does not determine which sectors contain BPS particles.
7. A BPS-quiver node is already an actual stable elementary BPS hypermultiplet. The automatic stability of its simple representation does not establish physical occupation.
8. Stability of general quiver representations tests composite bound states; it is not a post hoc test of whether the nodes are legitimate.
9. A BPS quiver requires a Coulomb vacuum, chamber, particle half-plane, finite positive basis, and completeness argument. Its superpotential and stability data are part of the presentation.
10. A physically identified finite-length heart with finitely many simples proves node stability and positive charge generation. An abstract categorical quiver is not automatically the physical BPS quiver.
11. Wall crossing can change the physical BPS spectrum; quiver mutation by itself changes the positive-basis description.
12. Smooth branch geometry controls tangent moduli, not necessarily every massless degree of freedom.

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